

VPN Server on Windows EC2 + RDP to Private EC2

AWS Practical - step by step with screenshots

What is inside

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1. Objective

Connect my personal Windows laptop to a **Windows EC2 instance that works as a VPN (RRAS) server** using the built-in Windows **PPTP** VPN. The laptop gets a VPN IP, and through the VPN server it reaches a **Windows EC2 in the private subnet** using only its **private IP (10.0.2.45)** over RDP. The private EC2 never gets a public IP.

Final path

```
My Windows Laptop (VPN IP 10.0.3.12)
  | PPTP VPN (TCP 1723 + GRE)
  v
Public Windows EC2 = VPN / RRAS server
  Private IP 10.0.1.163 | Public IP 44.199.227.64
  | routing + NAT inside the VPC
  v
Private Windows EC2 10.0.2.45
  | RDP TCP 3389
  v
Successful RDP connection
```

IP plan

Component	IP address	Purpose
VPN server - public IP	44.199.227.64	Laptop connects its VPN here
VPN server - private IP	10.0.1.163	RRAS server inside the VPC (public subnet)
VPN client pool	10.0.3.10 - 10.0.3.20	IPs given to VPN clients
Laptop VPN IP	10.0.3.12	Received after VPN connect
Private EC2	10.0.2.45	Final RDP destination (private subnet)
RDP port	3389	Windows Remote Desktop

Part 1 - Network setup (VPC, subnets, IGW, route table)

First a custom VPC is used with two subnets: a **public subnet** (VPN server lives here) and a **private subnet** (target server lives here).

Step 1.1 - Check the VPC and create two subnets

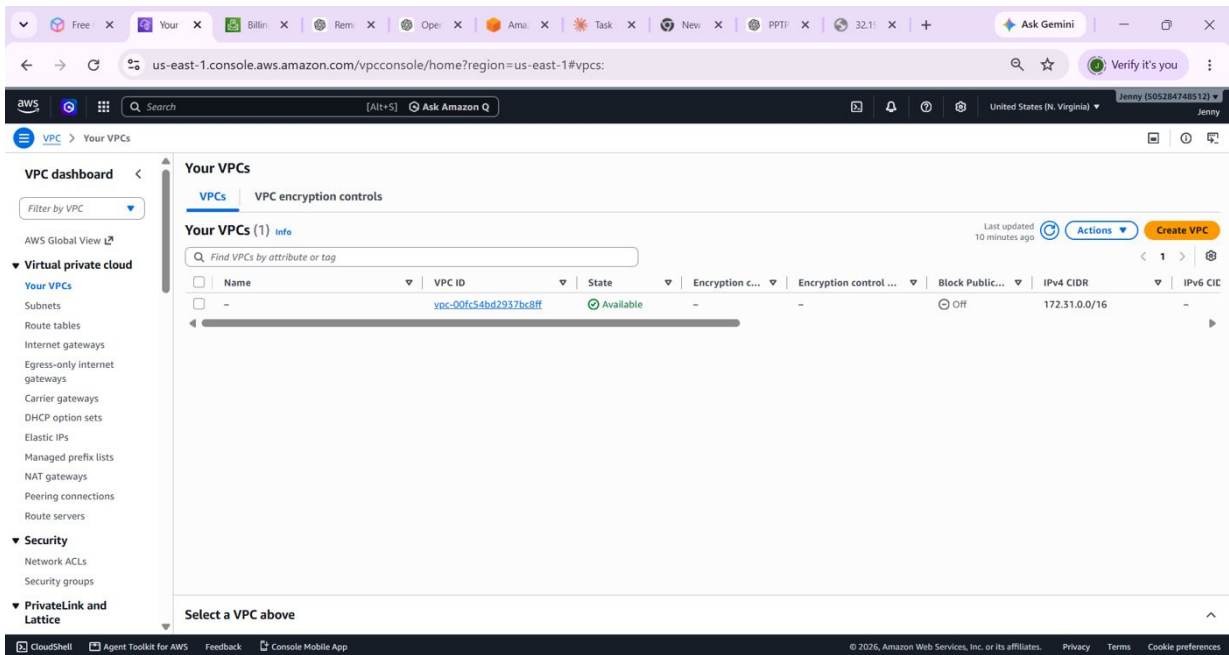


Fig 1. VPC dashboard - the custom VPC that will be used.

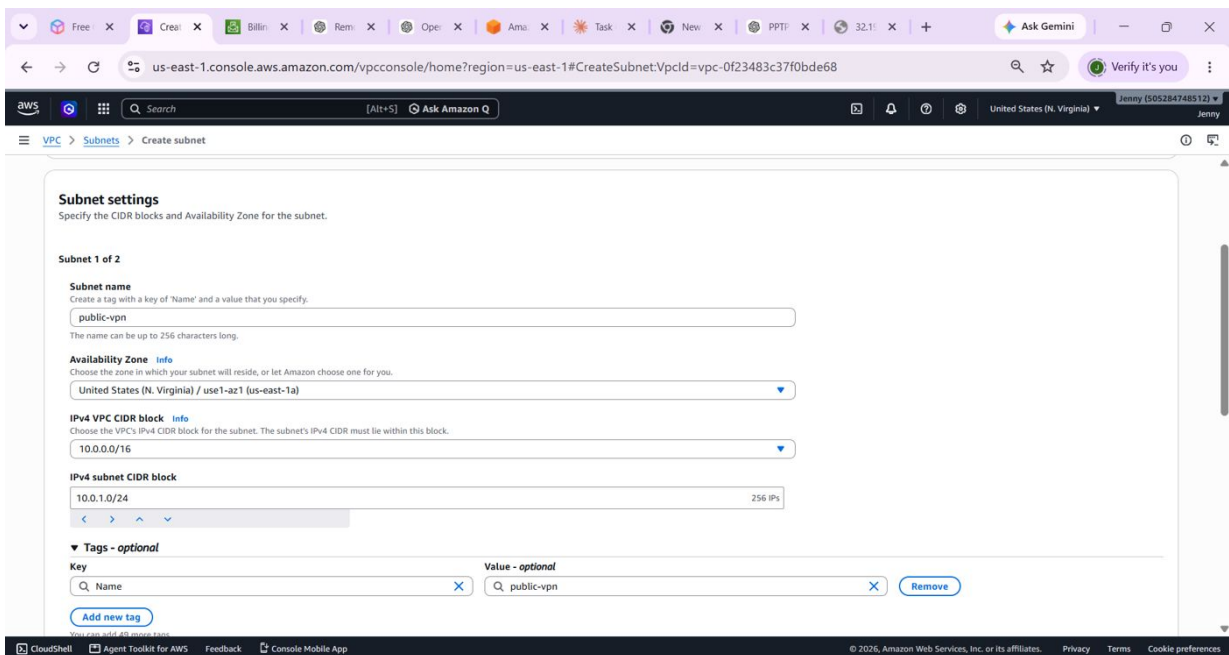


Fig 2. Create subnet - first subnet (public-sub) with its CIDR block (10.0.1.0/24).

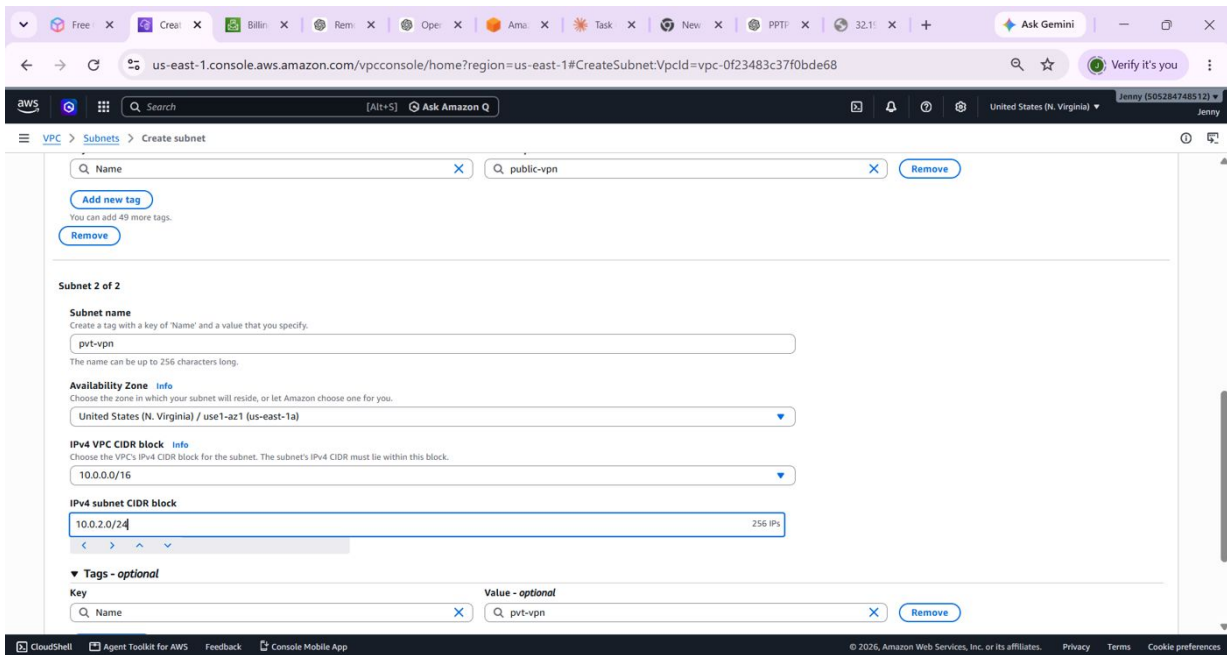


Fig 3. Second subnet (pvt-sub) added in the same wizard (10.0.2.0/24).

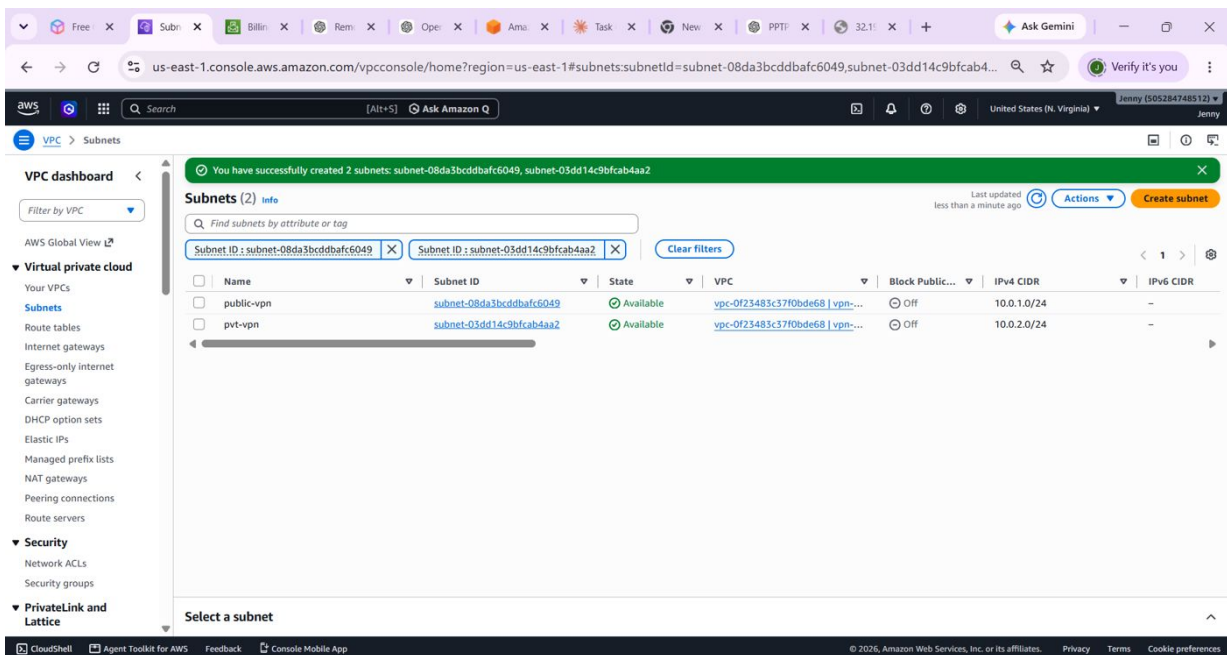


Fig 4. Both subnets created successfully.

Step 1.2 - Create and attach the Internet Gateway

The Internet Gateway (IGW) gives the public subnet a path to the internet. Without it the laptop cannot reach the VPN server.

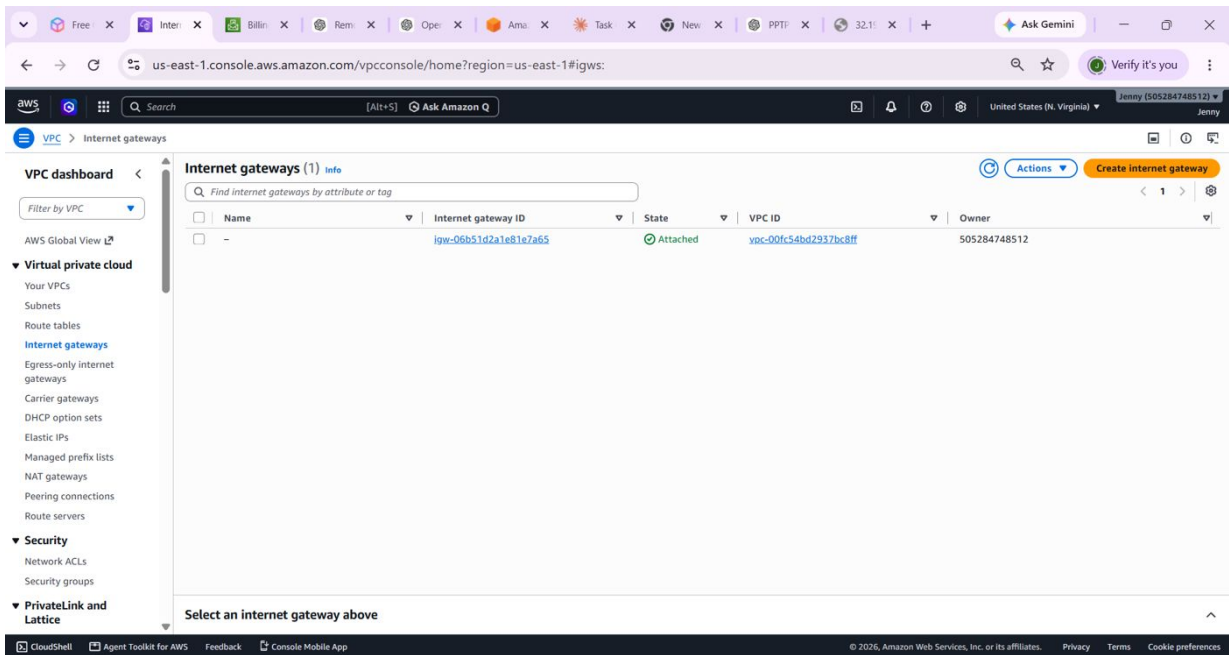


Fig 5. Internet gateways list before creating a new one.

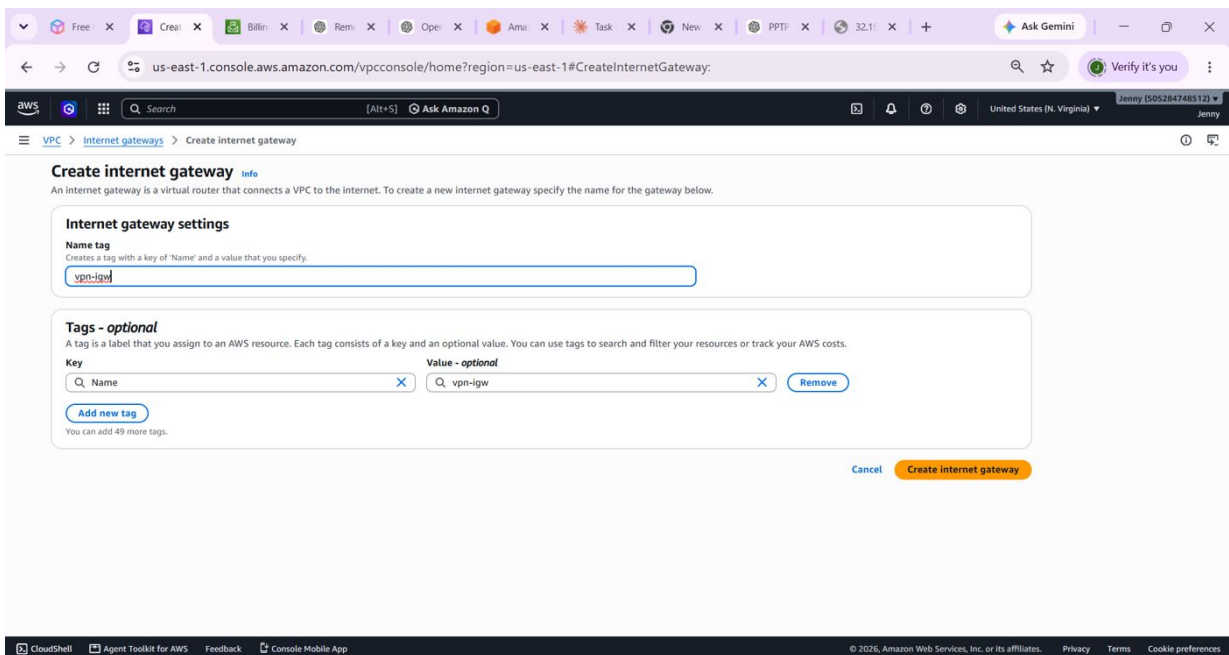


Fig 6. Create internet gateway - give it a name (vpn-igw).

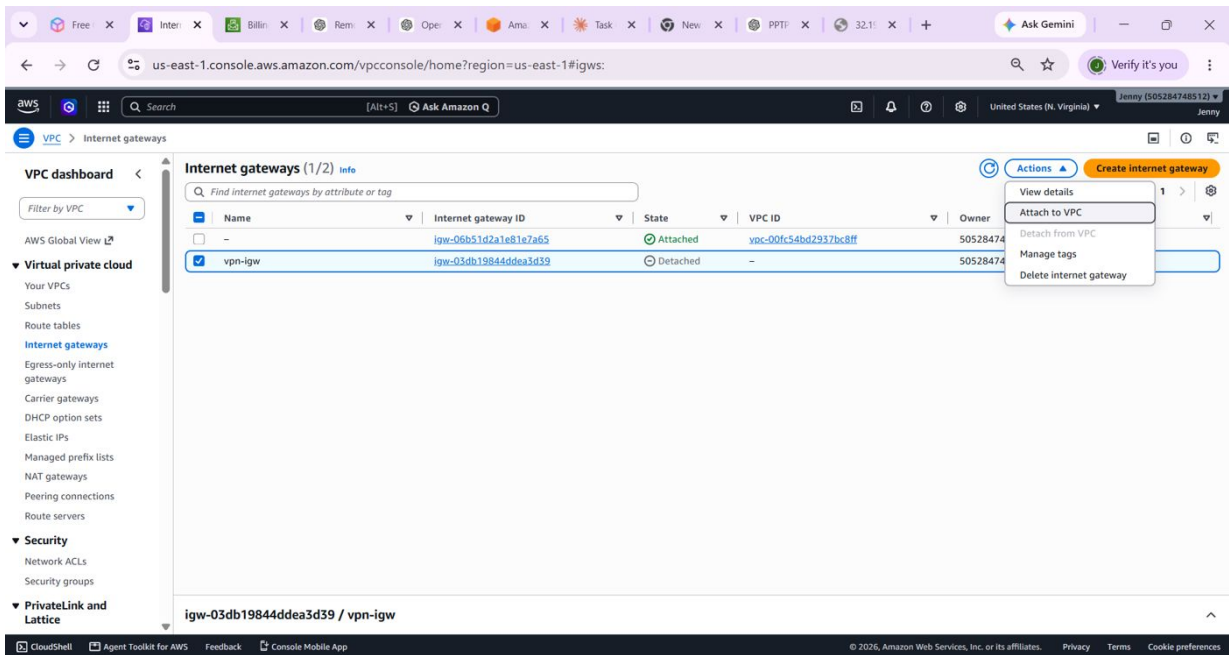


Fig 7. Actions -> Attach to VPC.

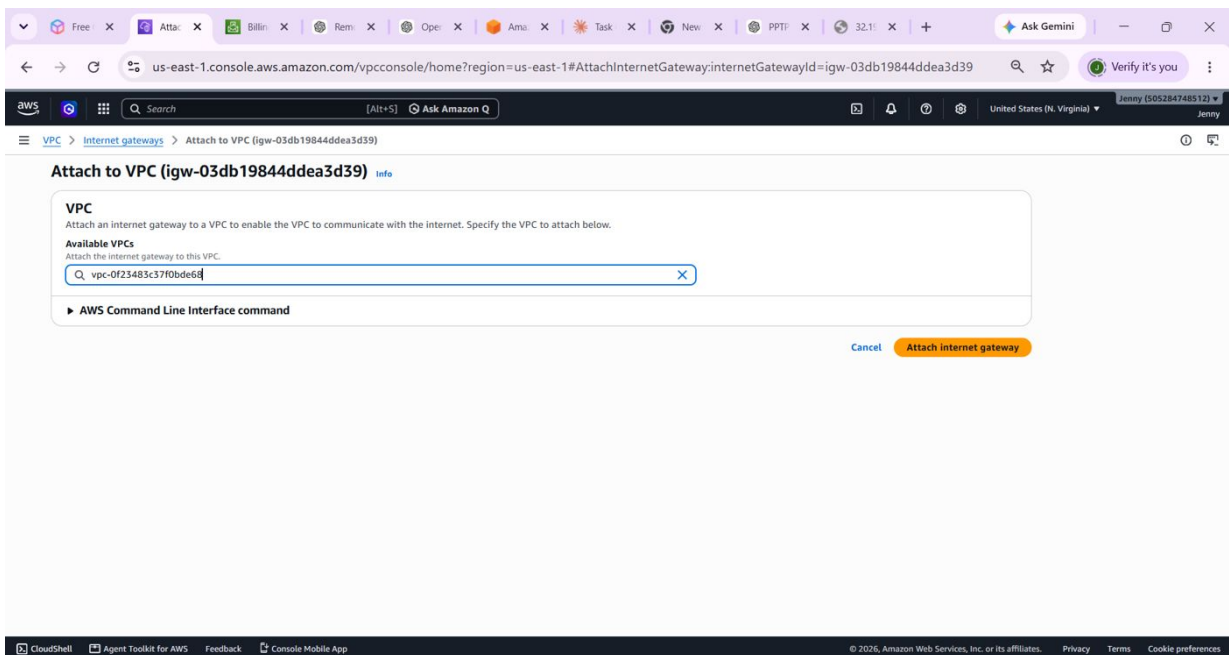


Fig 8. Select the VPC and click Attach internet gateway.

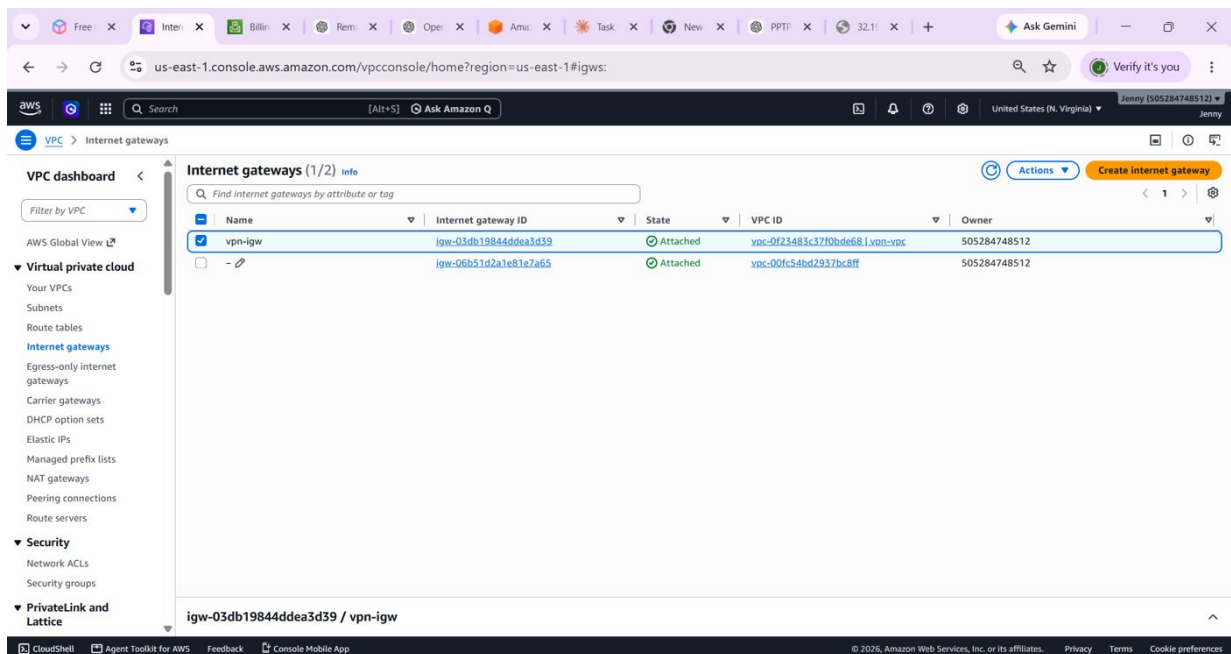


Fig 9. IGW state shows Attached.

Step 1.3 - Public route table

A route table with **0.0.0.0/0** -> **Internet Gateway** is what makes a subnet 'public'.

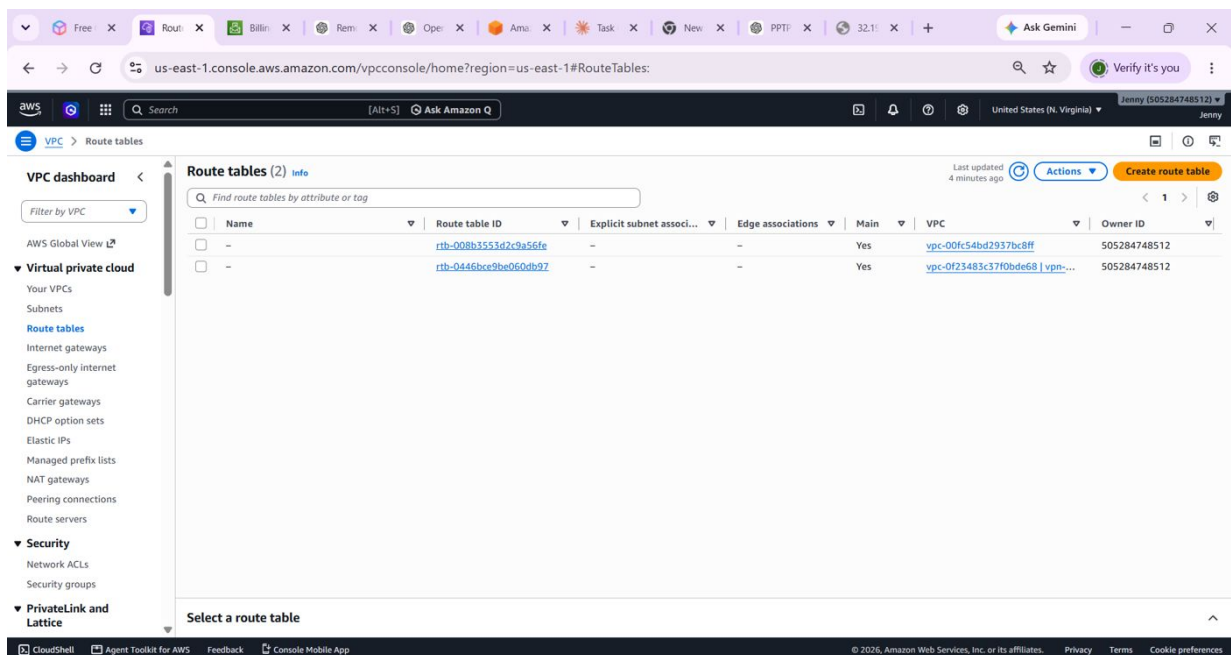


Fig 10. Route tables list.

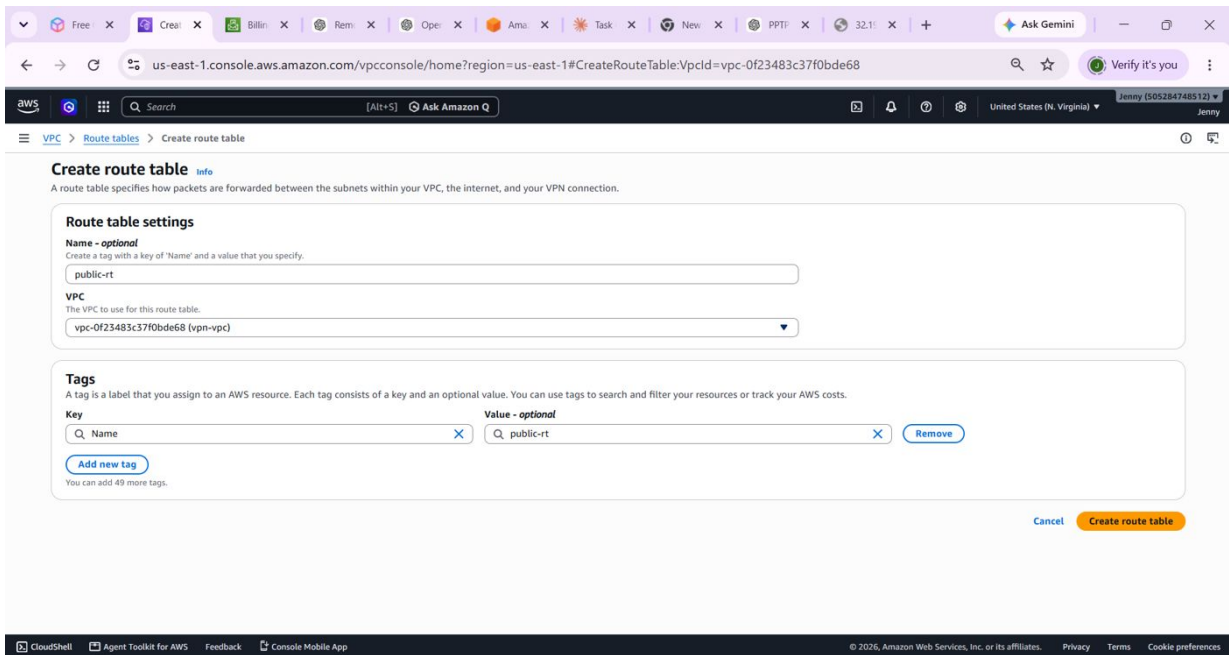


Fig 11. Create route table (public-rt) inside the VPC.

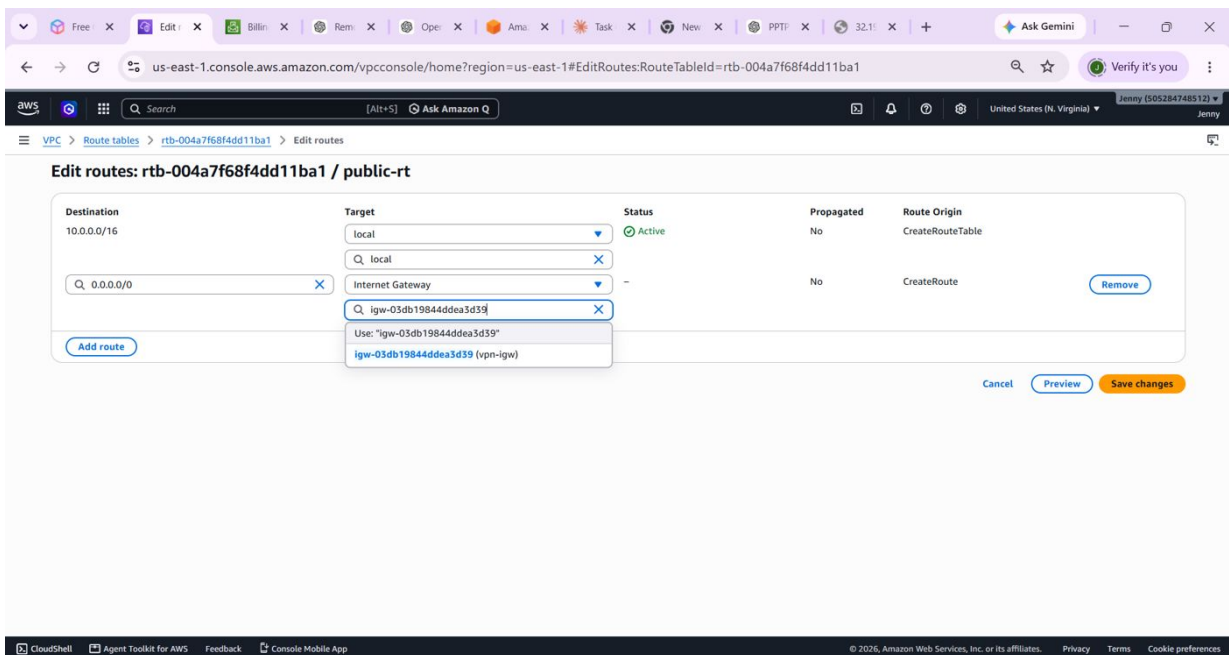


Fig 12. Edit routes - add 0.0.0.0/0 with target = Internet Gateway.

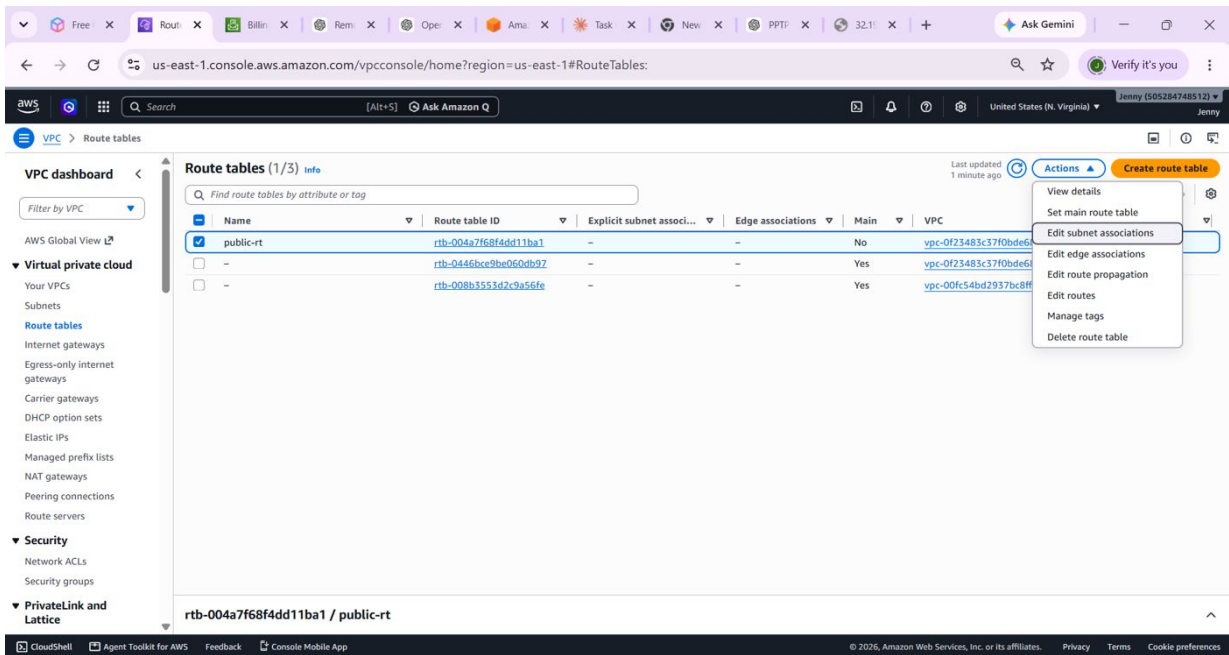


Fig 13. Actions -> Edit subnet associations.

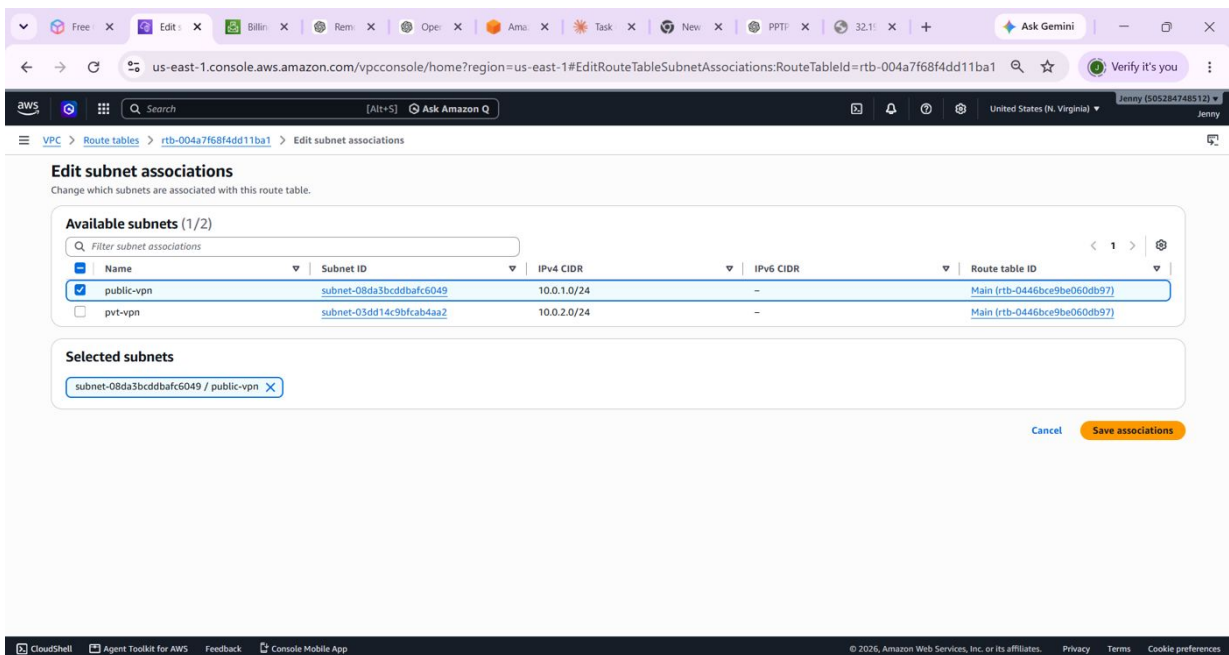


Fig 14. Associate only the public subnet with this route table.

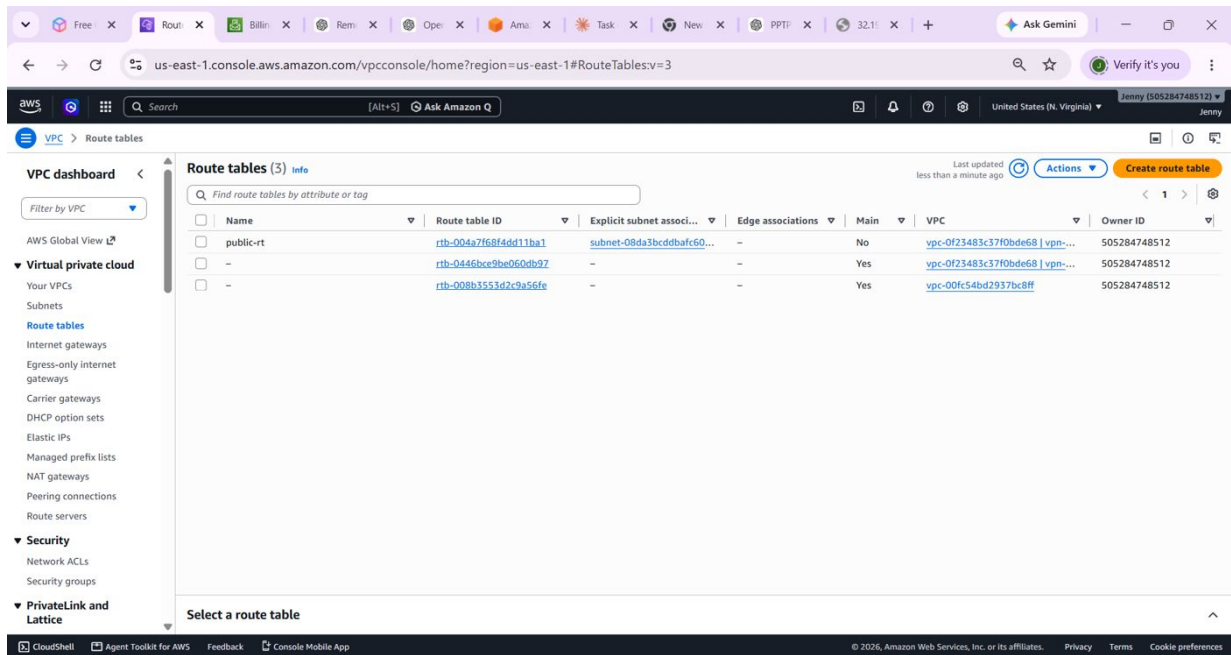


Fig 15. Route table ready.

Step 1.4 - Auto-assign public IP on the public subnet

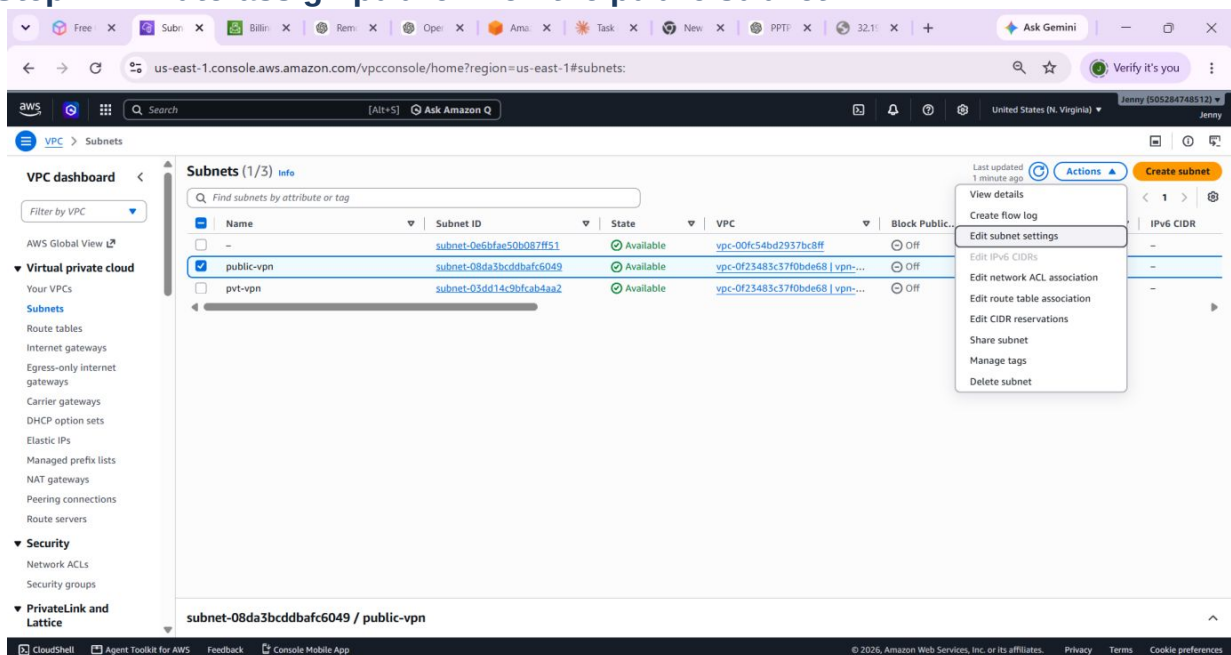


Fig 16. Subnets list - select the public subnet -> Actions -> Edit subnet settings.

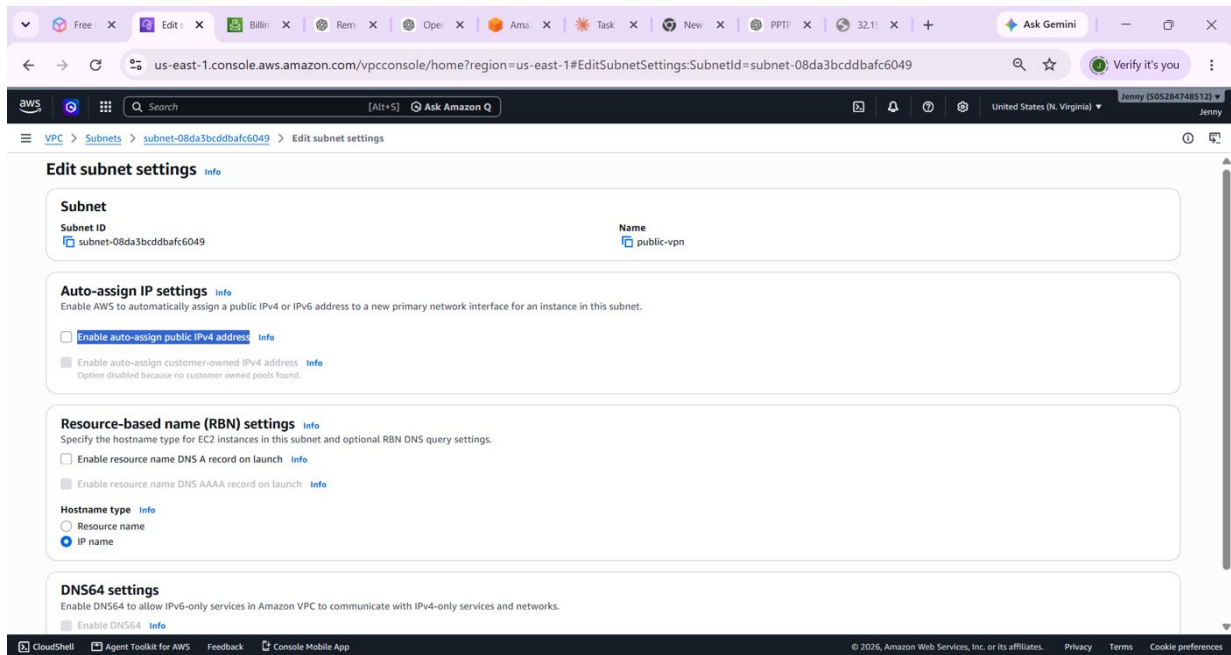


Fig 17. Enable 'auto-assign public IPv4 address' so the VPN server gets a public IP.

Part 2 - Launch the servers and first RDP

Step 2.1 - Launch the Public Windows EC2 (VPN server)

Launch a **Windows Server 2025 Base** instance in the public subnet, with a key pair, and a security group that allows RDP.

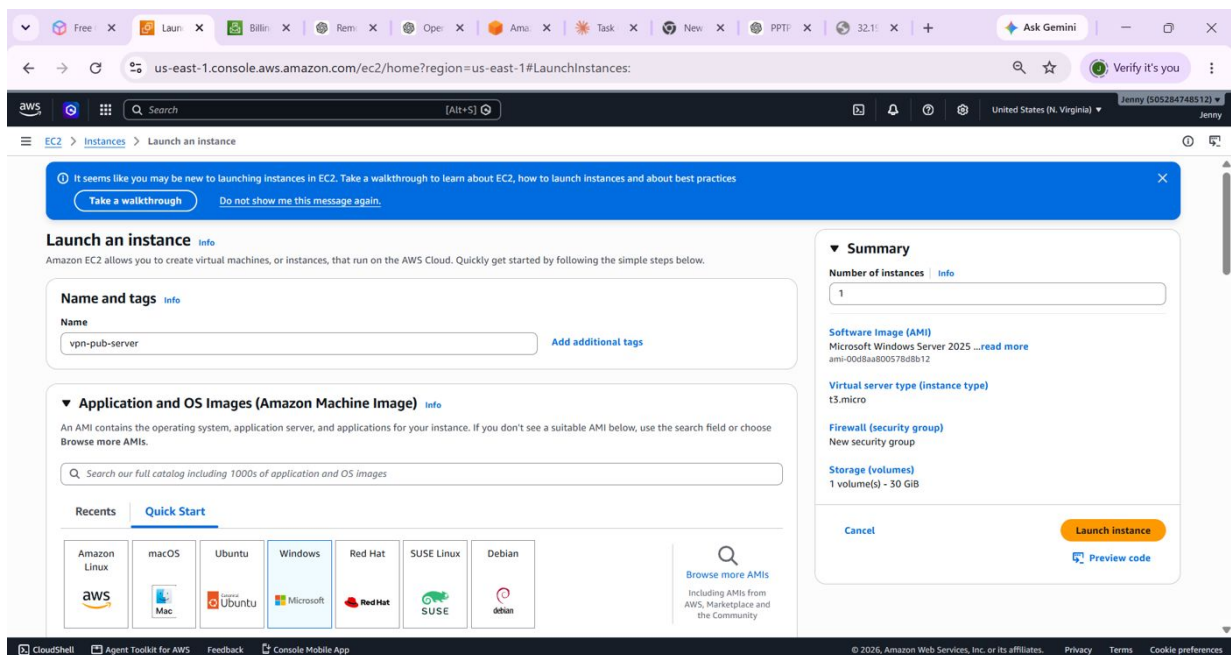


Fig 18. Launch instance - name and Windows AMI selection.

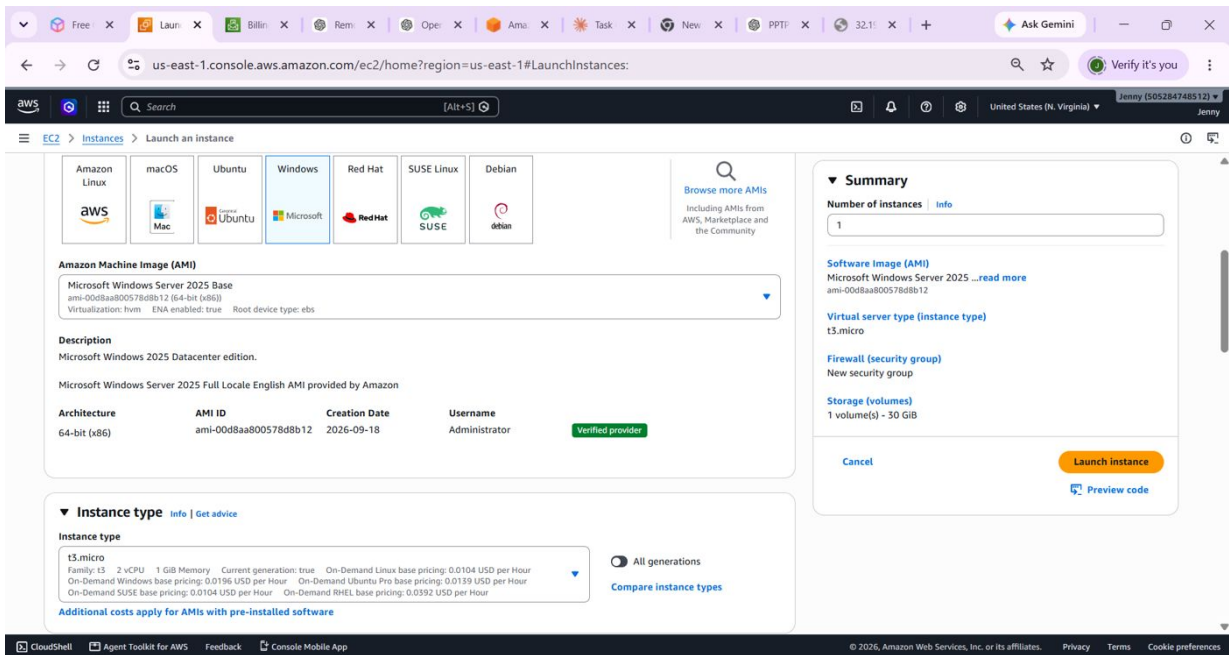


Fig 19. Windows Server AMI details (Quick Start -> Windows).

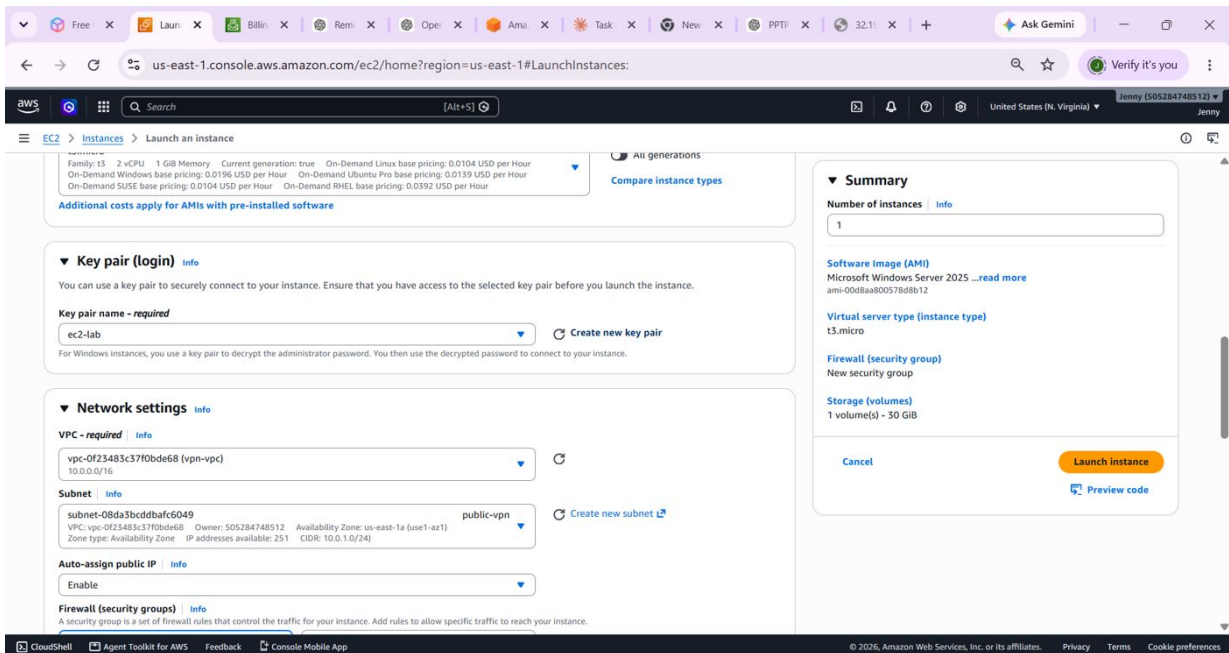


Fig 20. Instance type, key pair and network settings.

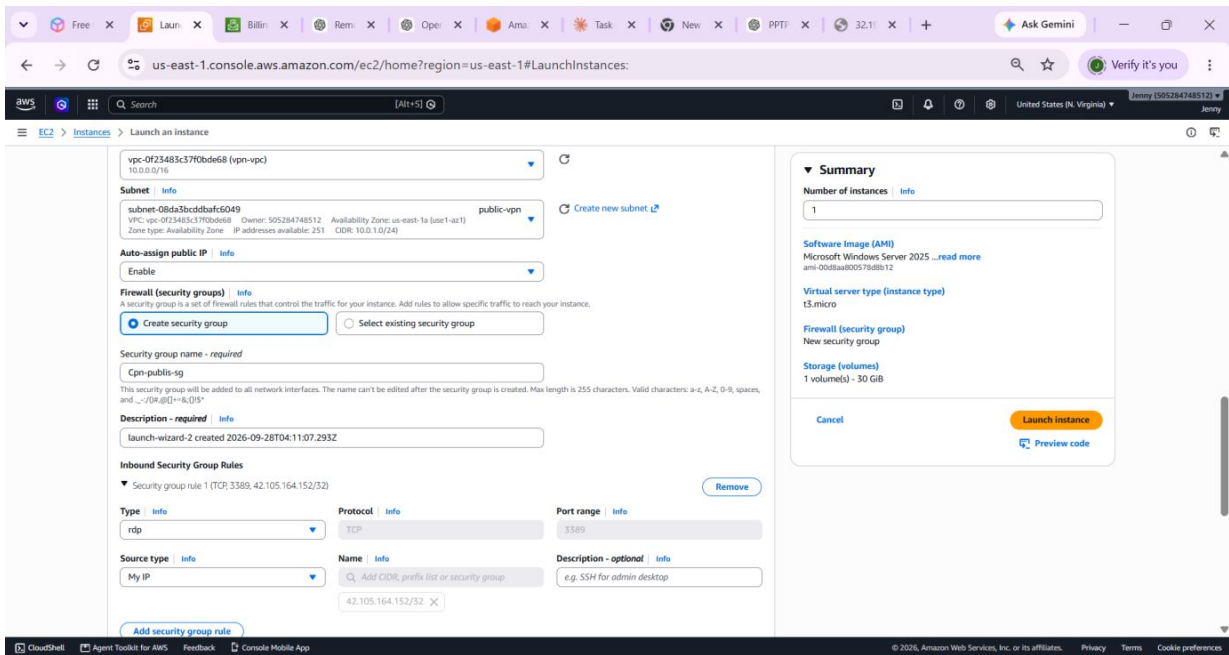


Fig 21. Network settings - public subnet selected, auto-assign public IP enabled.

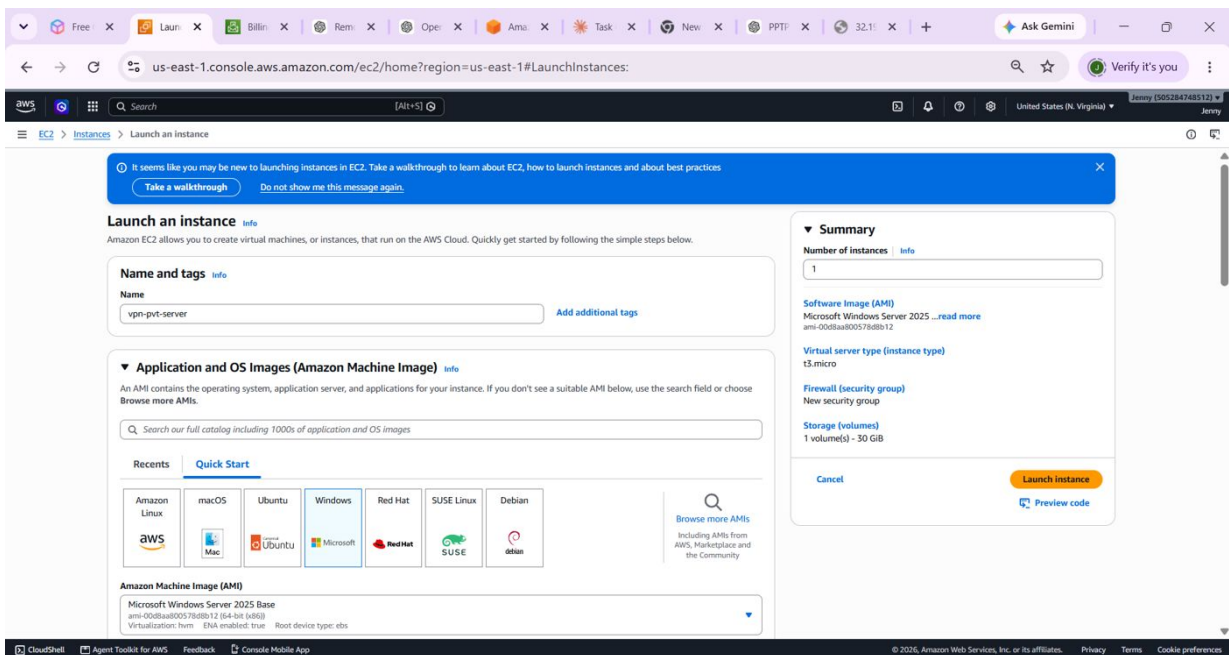


Fig 22. Second instance launch - name the private server.

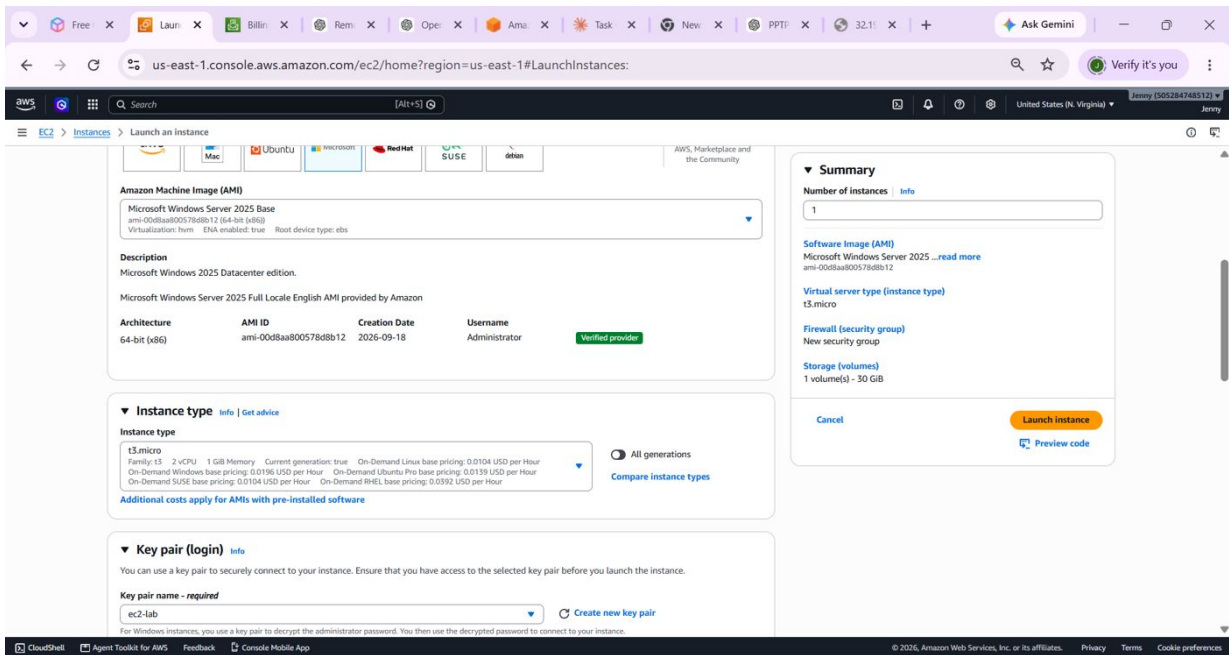


Fig 23. Private server AMI (Windows Server) selected.

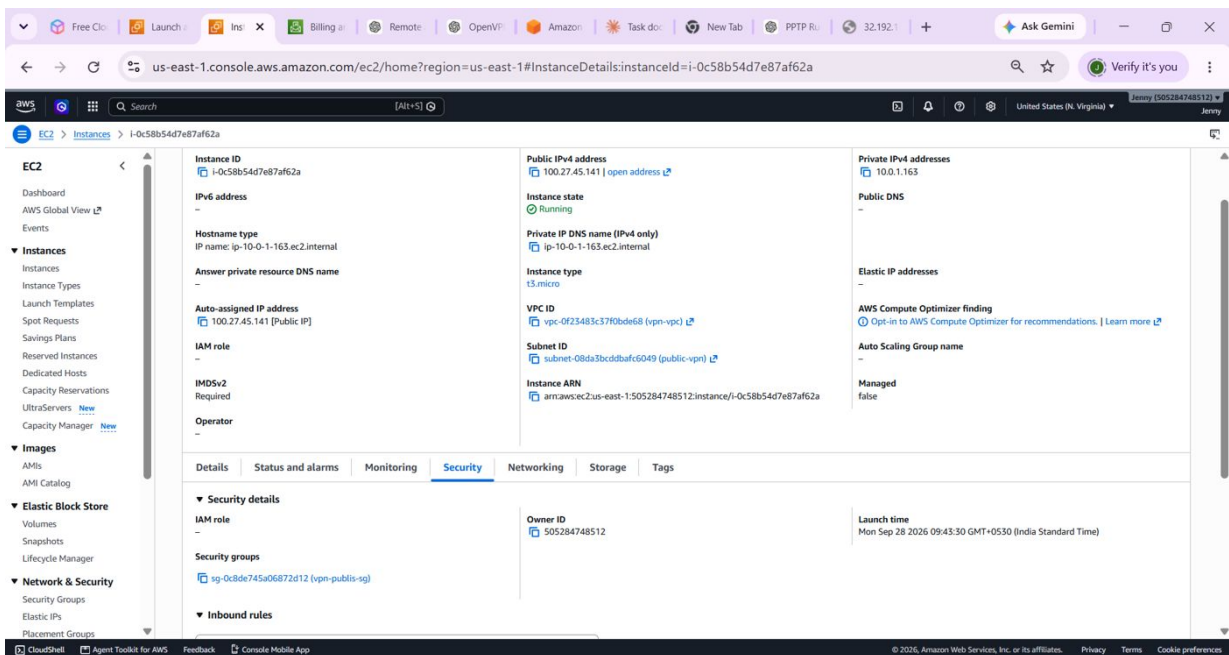


Fig 24. Instance details after launch - shows the private IP address and security tab.

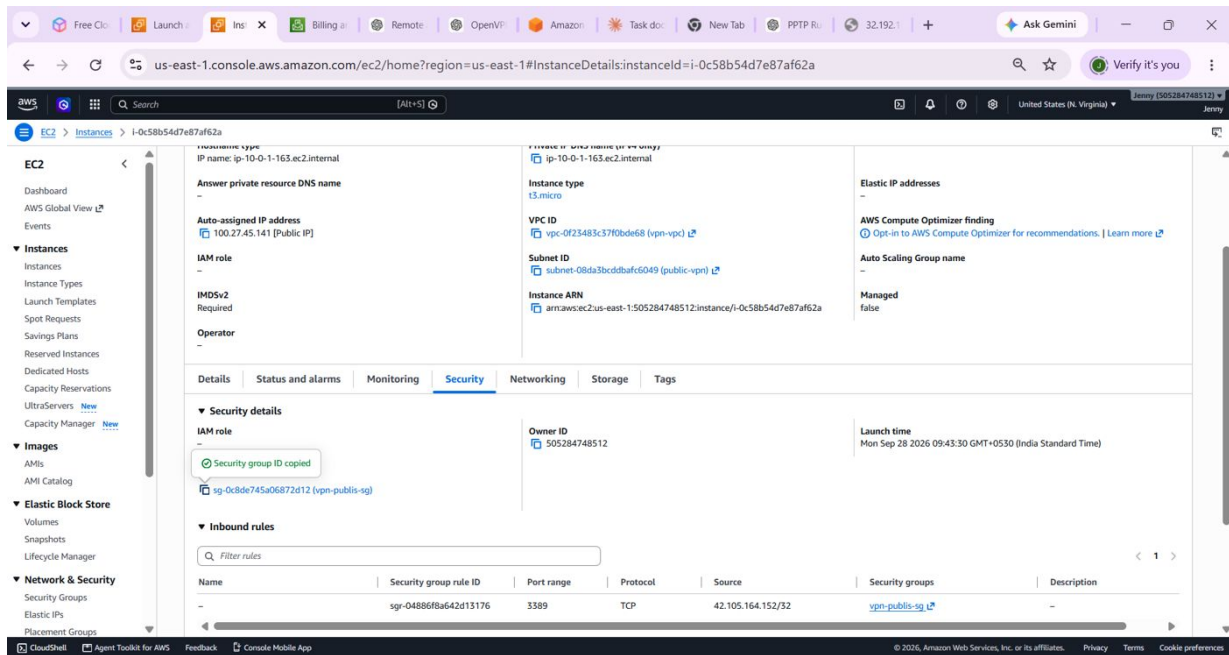


Fig 25. Security tab - inbound rules of the instance.

Step 2.2 - Private EC2 security group

The private EC2 only needs to accept traffic from inside the VPC (from the VPN server security group). It gets no public IP.

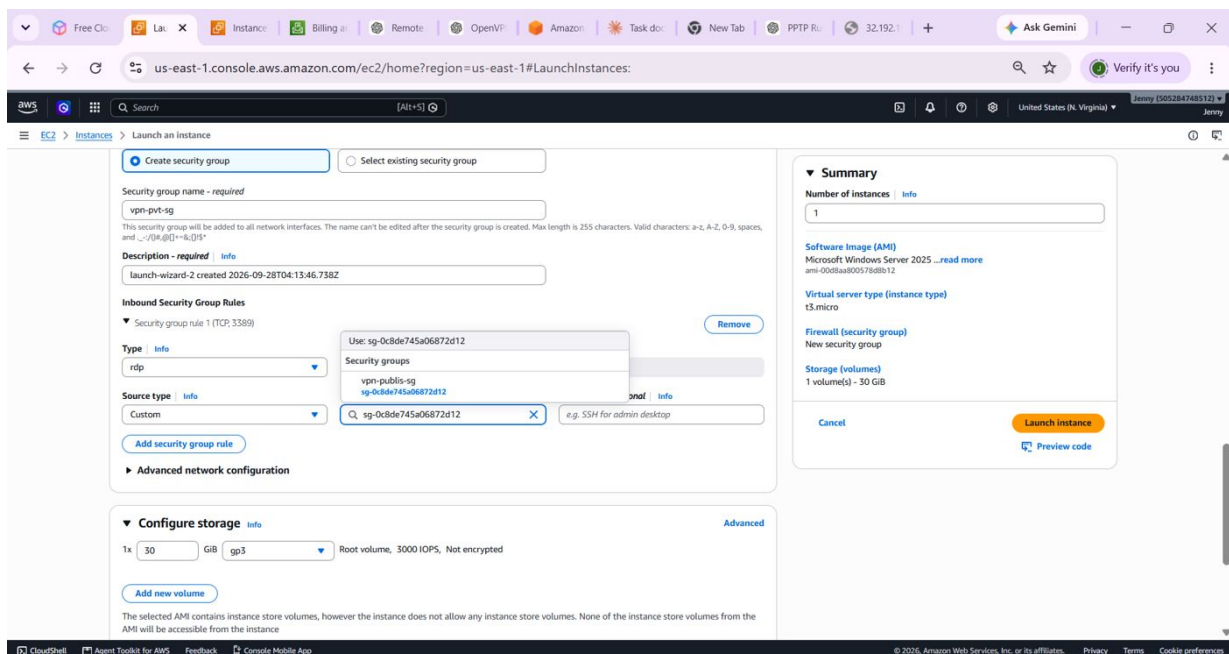


Fig 26. Private server security group - RDP allowed from the VPN server's security group.

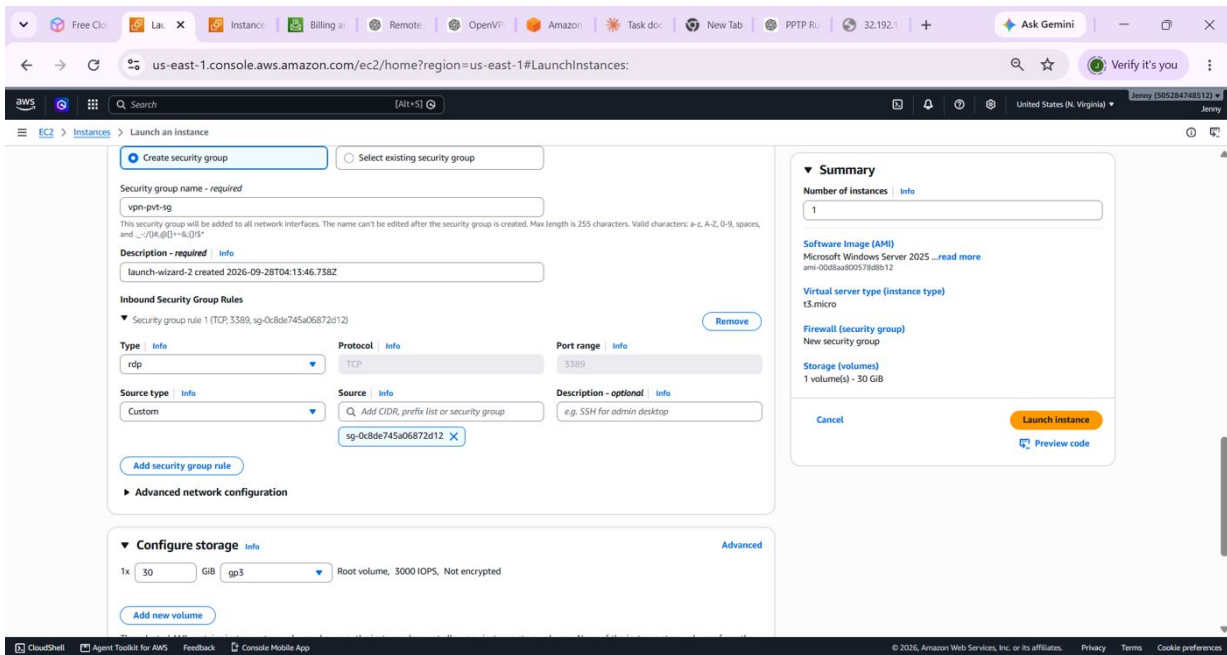


Fig 27. Security group and storage confirmed before launching.

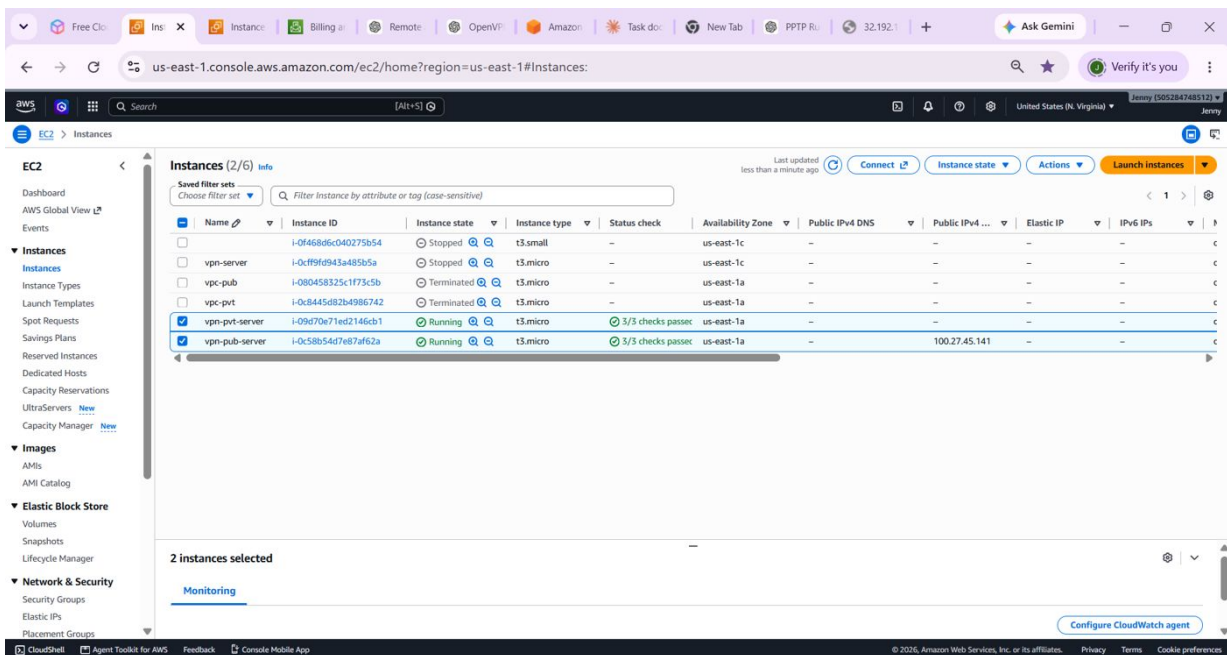


Fig 28. Instances list - both servers running (public VPN server and private server).

Step 2.3 - First RDP into the Public EC2

Only the public EC2 is reached directly over the internet (once), just to configure it as the VPN server.

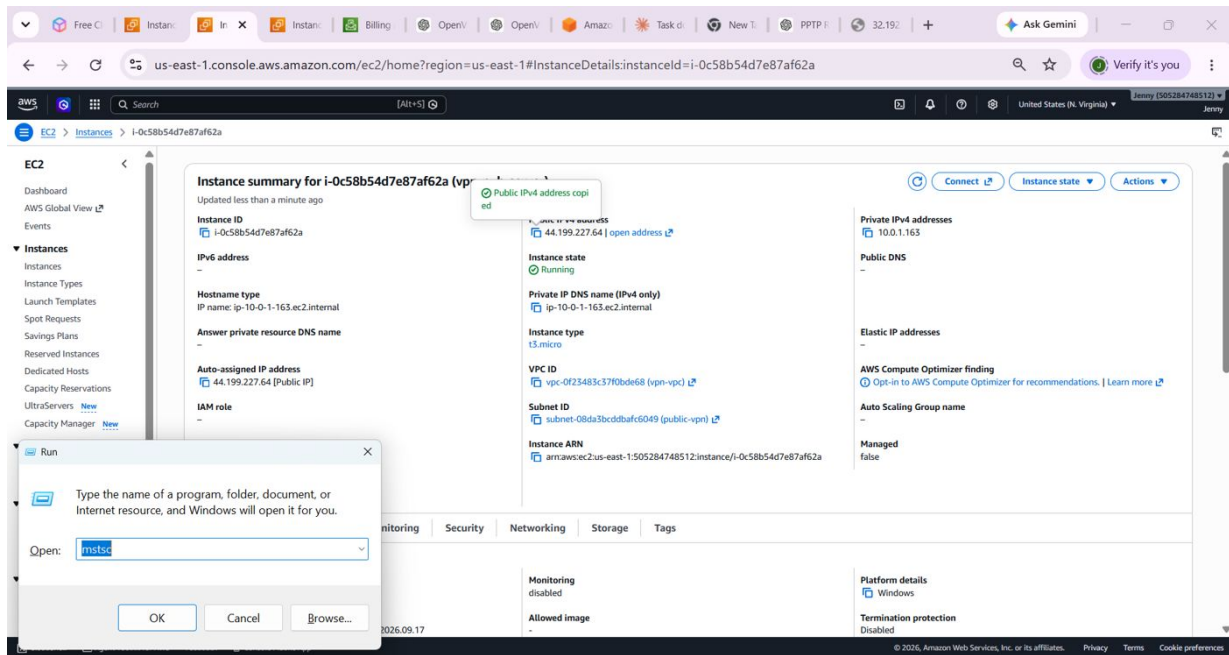


Fig 29. Press Win + R and type mstsc to open Remote Desktop.

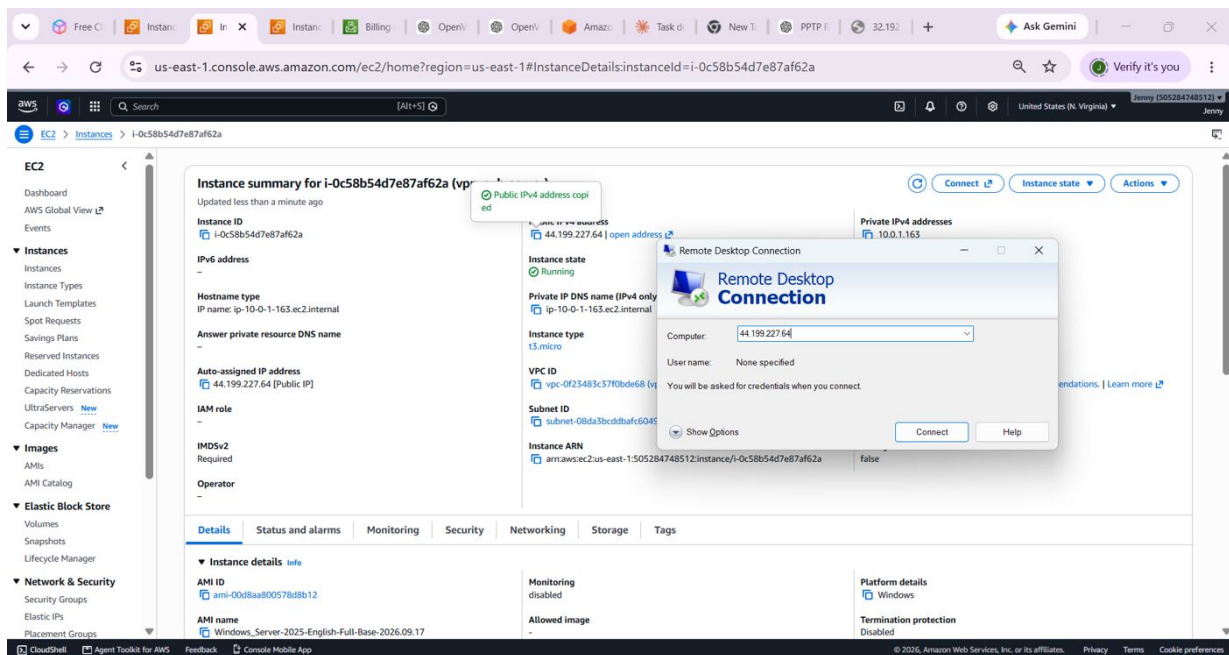


Fig 30. Connect to the public IP of the VPN server. Windows password is decrypted using the key pair.

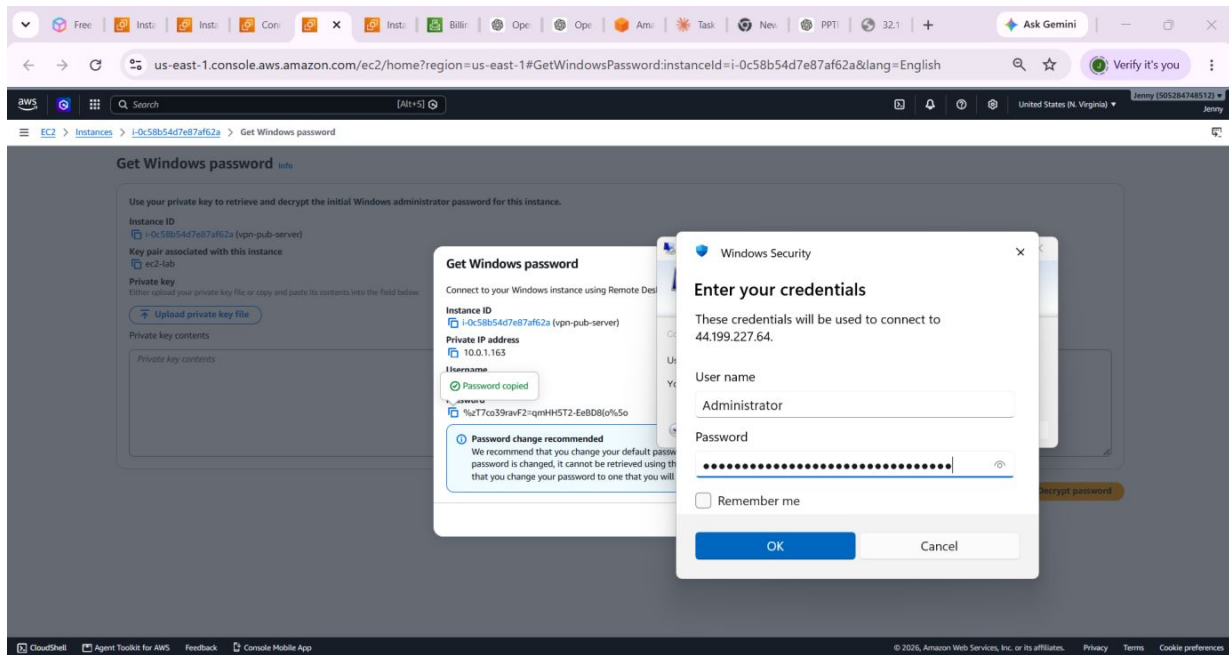


Fig 31. Enter Administrator credentials.

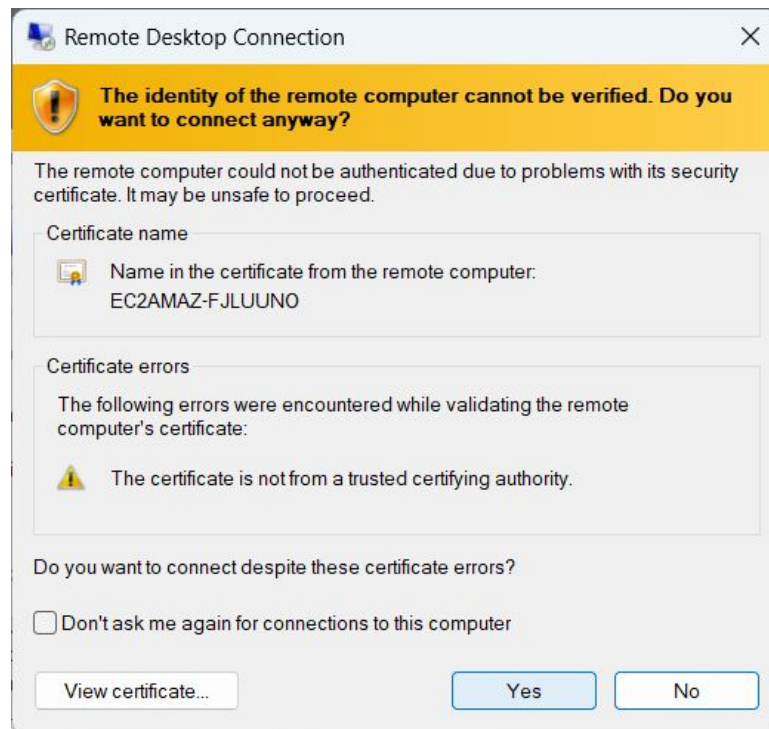


Fig 32. Accept the certificate warning - the RDP session opens.

Part 3 - Install Remote Access and configure RRAS

Step 3.1 - Add Roles and Features

Inside the VPN server: **Server Manager -> Manage -> Add Roles and Features**. Choose *Role-based or feature-based installation*, select this server, and tick **Remote Access**.

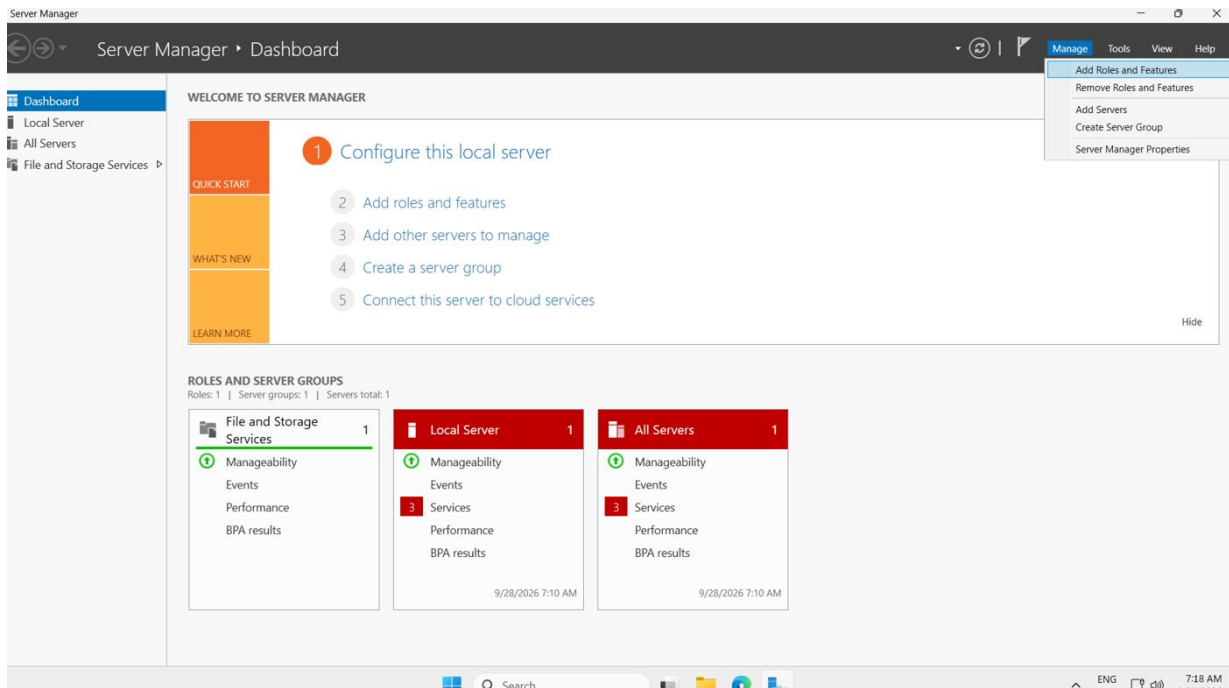


Fig 33. Server Manager dashboard - Manage -> Add Roles and Features.

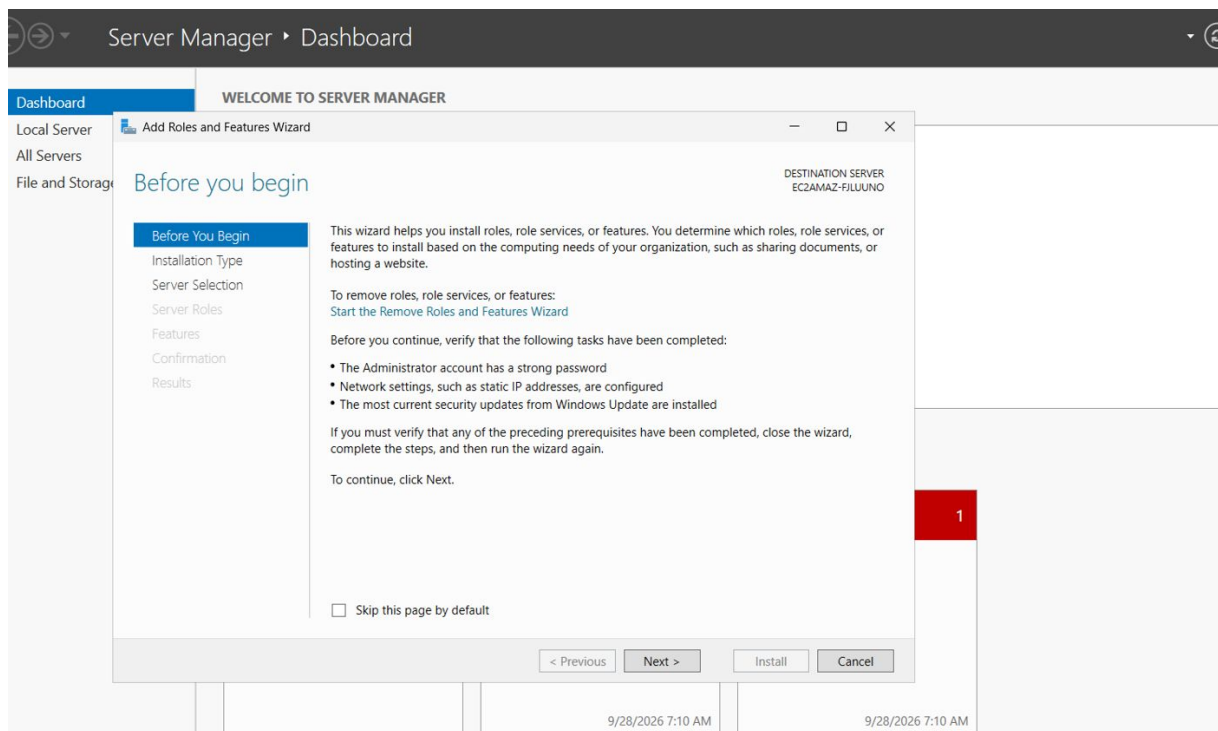


Fig 34. Before you begin - click Next.

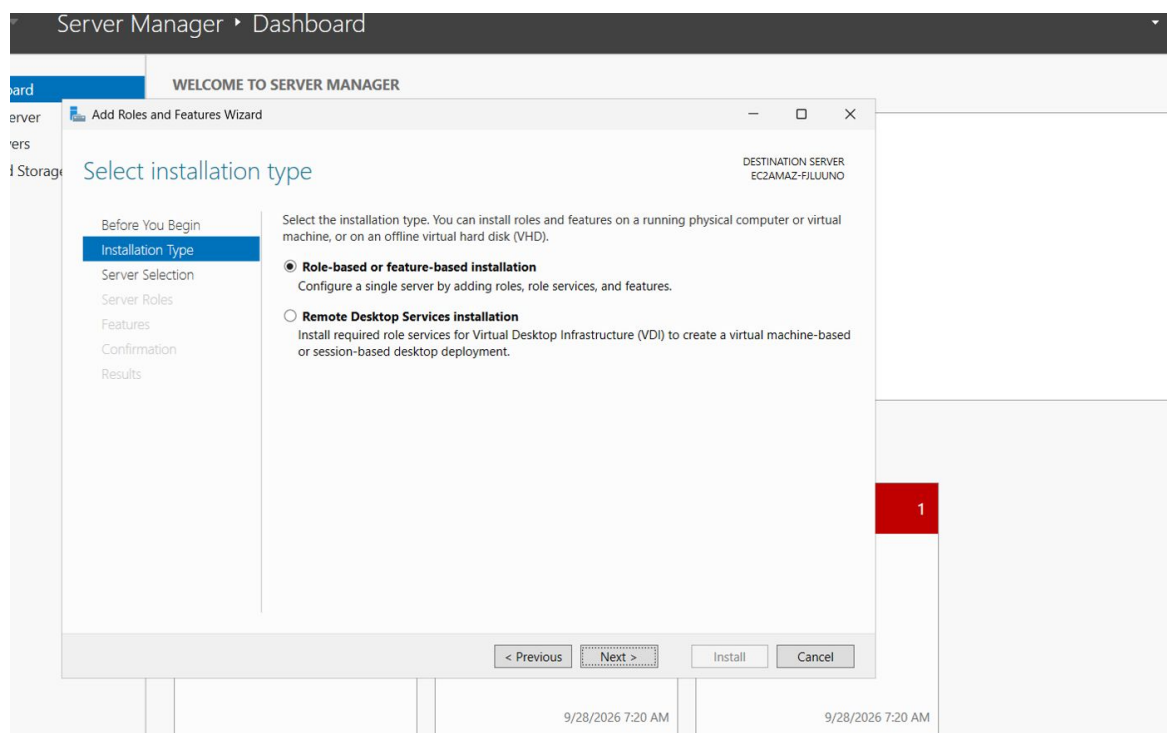


Fig 35. Installation type: Role-based or feature-based installation.

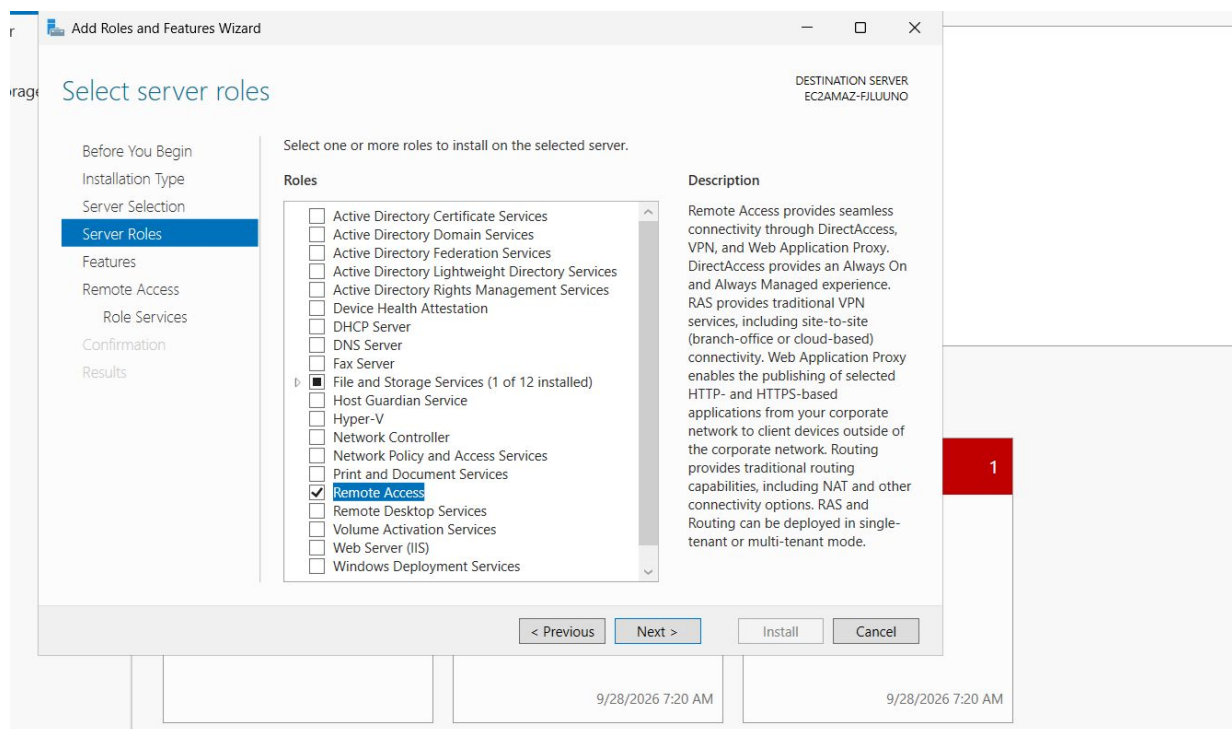


Fig 36. Server roles: Remote Access ticked.

On the next pages do not select any extra Features. Under Role Services tick **DirectAccess** and **VPN (RAS)** and **Routing** (both are required). Accept 'Add Features' and click Install.

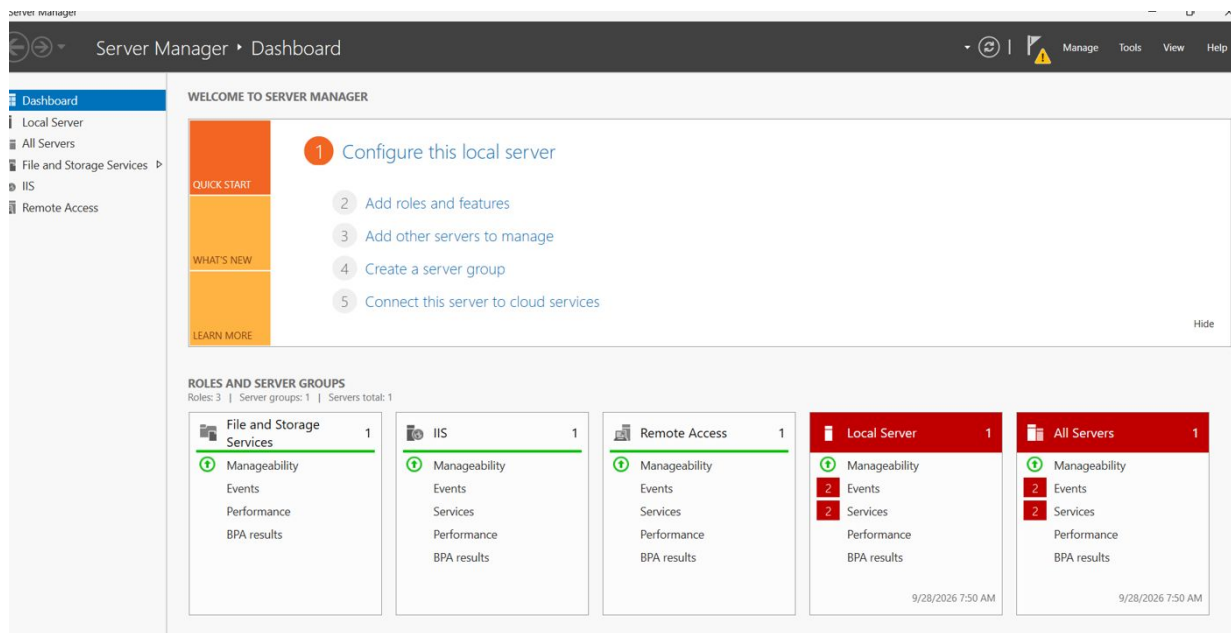


Fig 37. After installation Remote Access appears in the Server Manager dashboard.

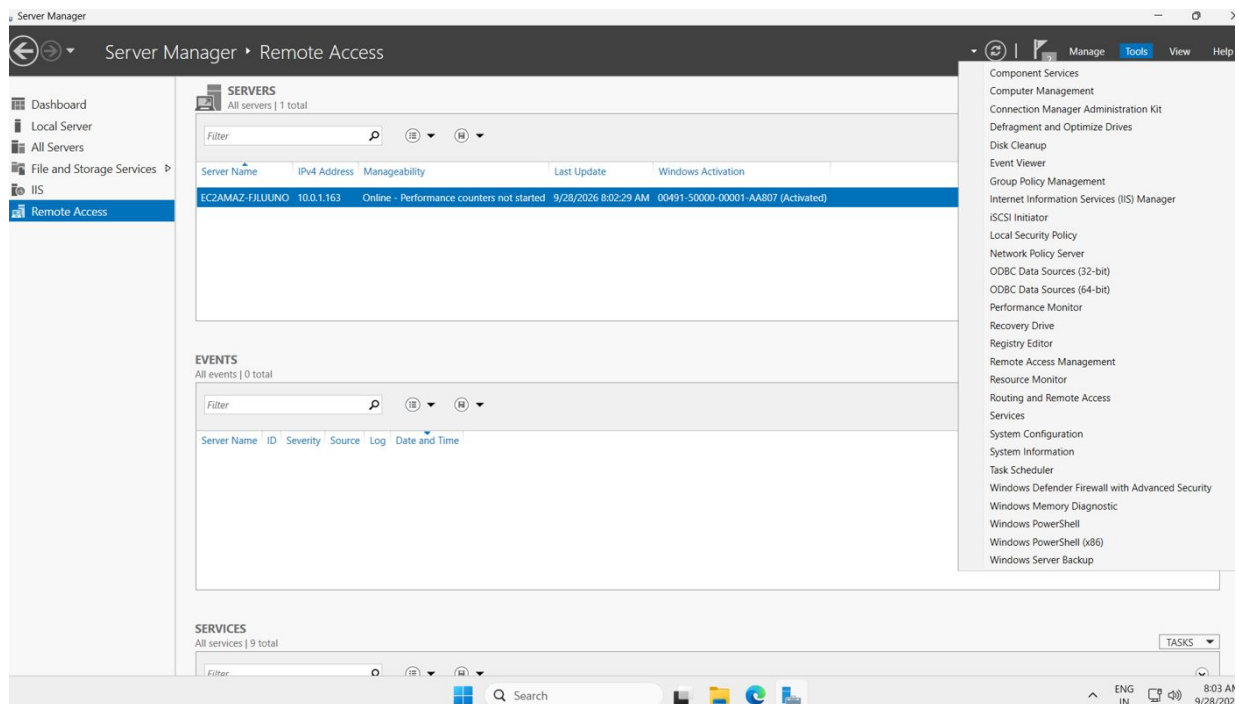


Fig 38. Remote Access page - open Tools -> Routing and Remote Access.

Step 3.2 - Configure and enable Routing and Remote Access

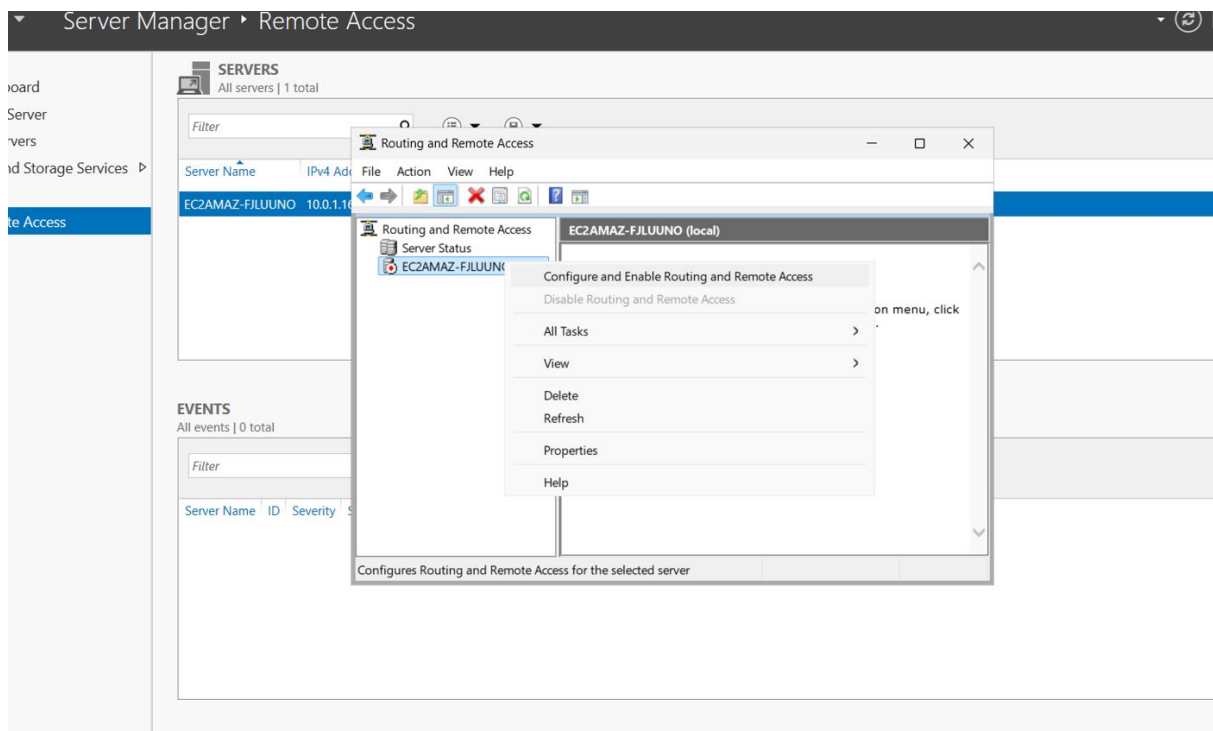


Fig 39. Right-click the server -> Configure and Enable Routing and Remote Access.

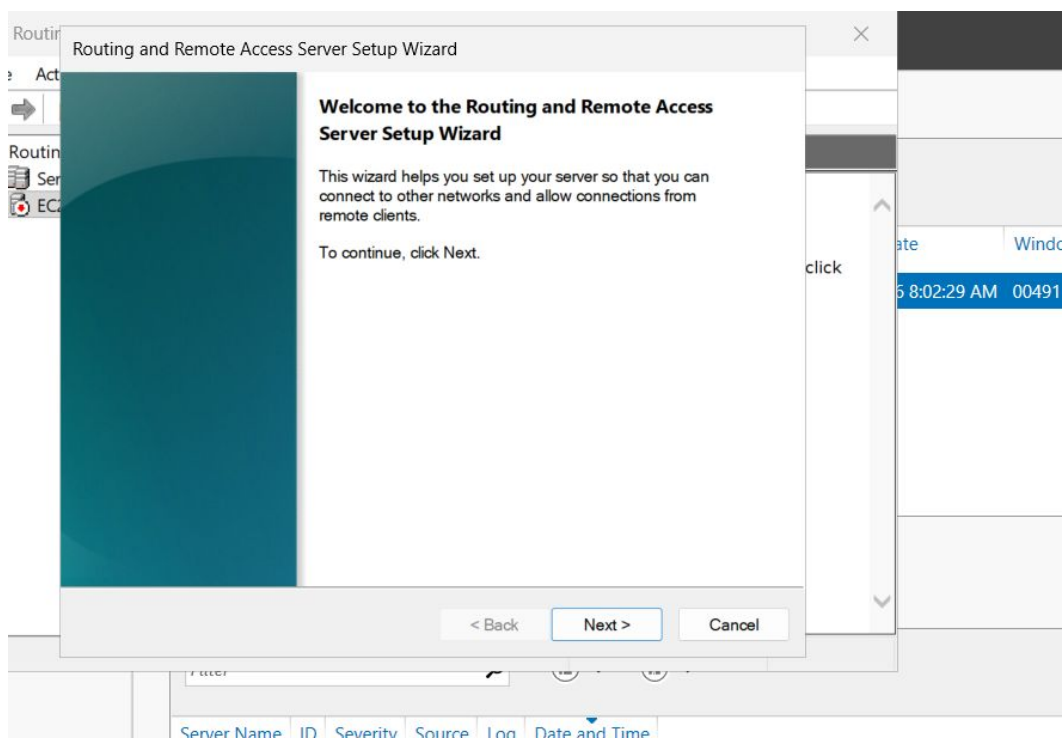


Fig 40. Setup wizard welcome page - Next.

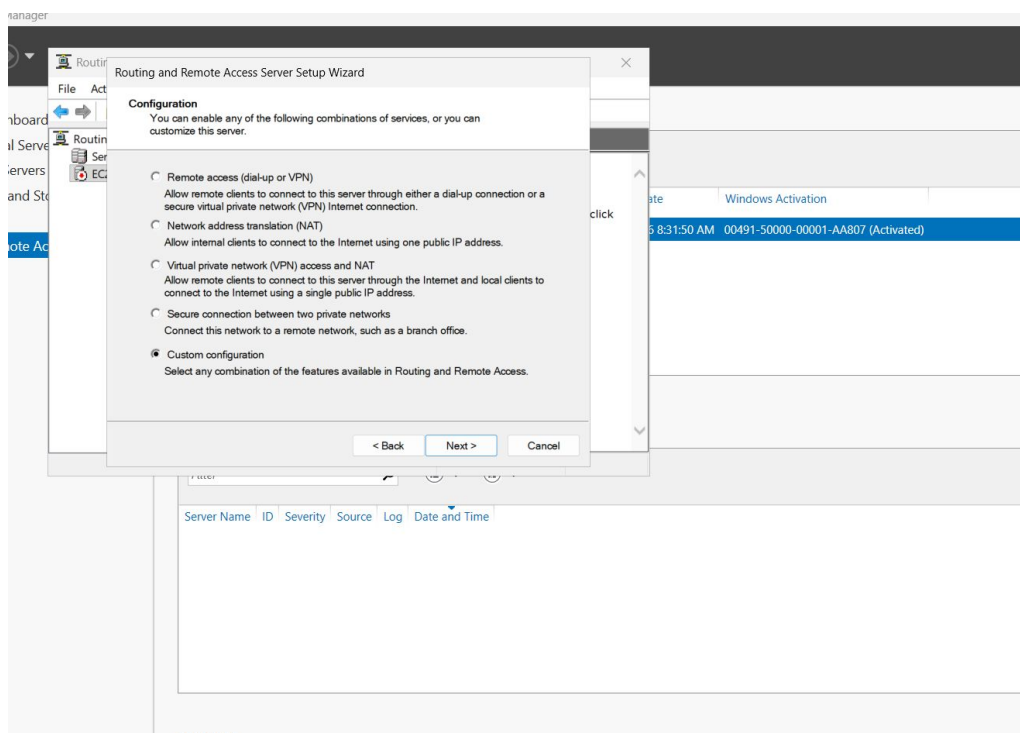


Fig 41. Configuration: choose Custom configuration.

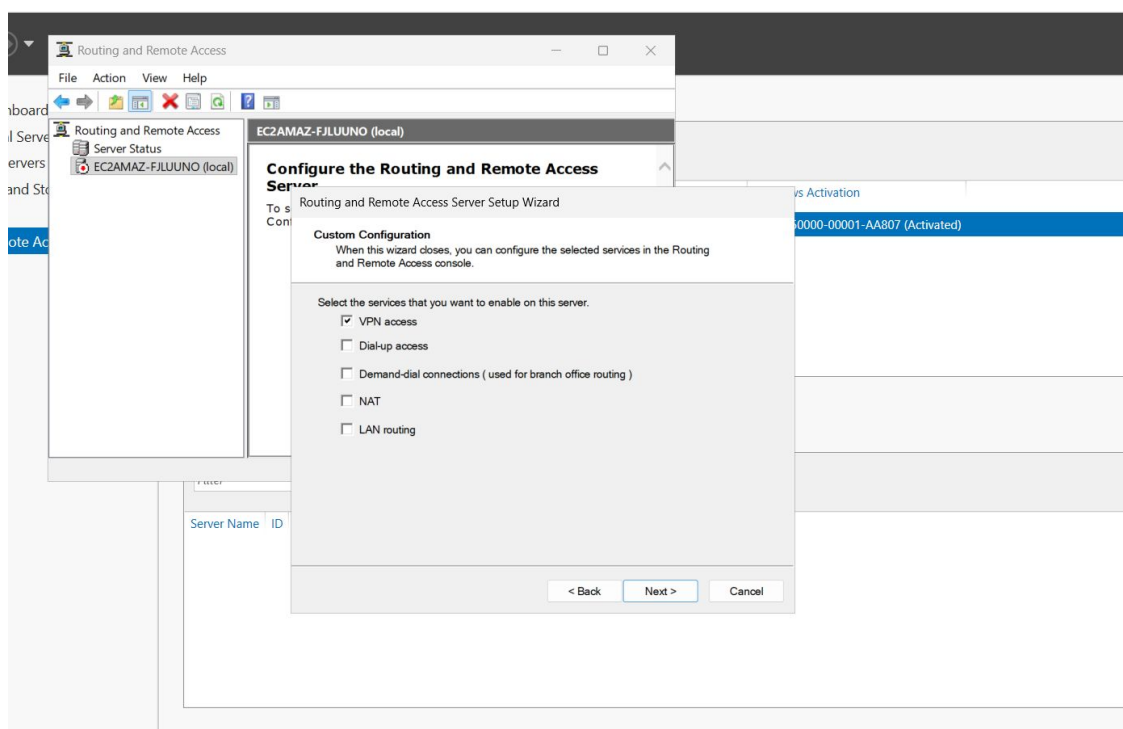


Fig 42. Custom configuration: tick VPN access only.

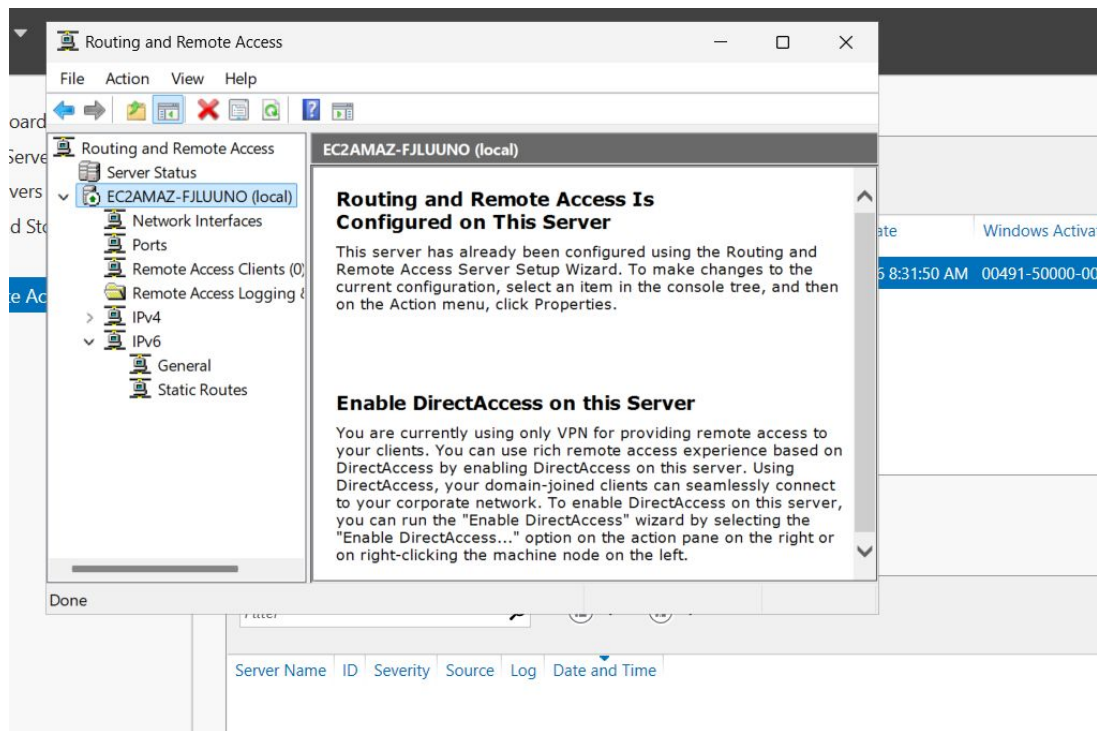


Fig 43. RRAS is configured on the server. Start the service.

Step 3.3 - VPN client IP pool

Right-click server -> Properties -> **IPv4** tab -> Static address pool. Range used: **10.0.3.10 to 10.0.3.20**. Every VPN client will get one IP from this pool (laptop got 10.0.3.12).

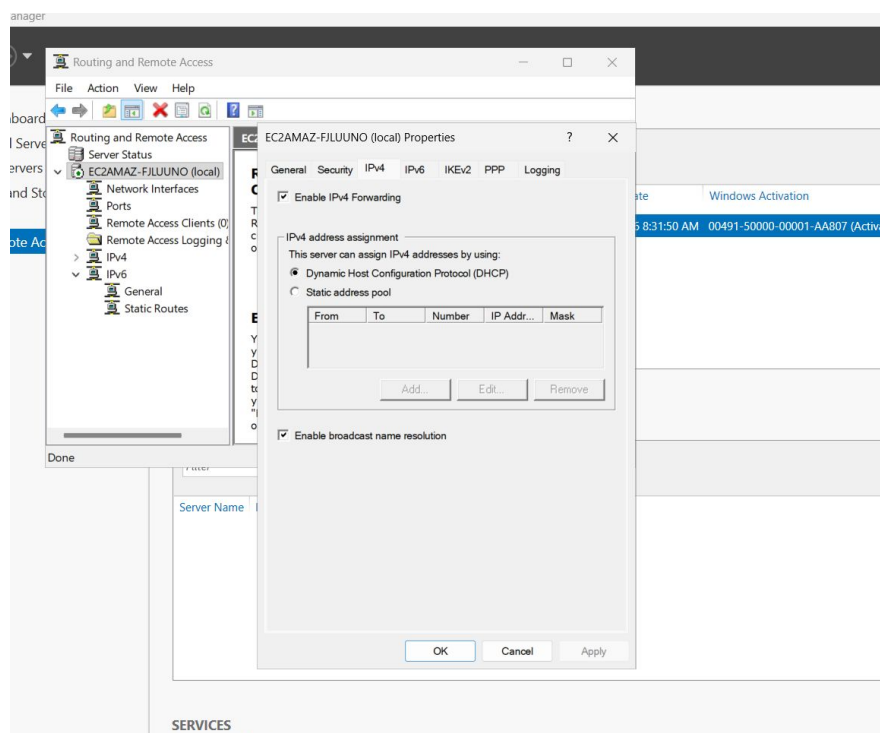


Fig 44. Server properties -> IPv4 tab (Enable IPv4 Forwarding ticked).

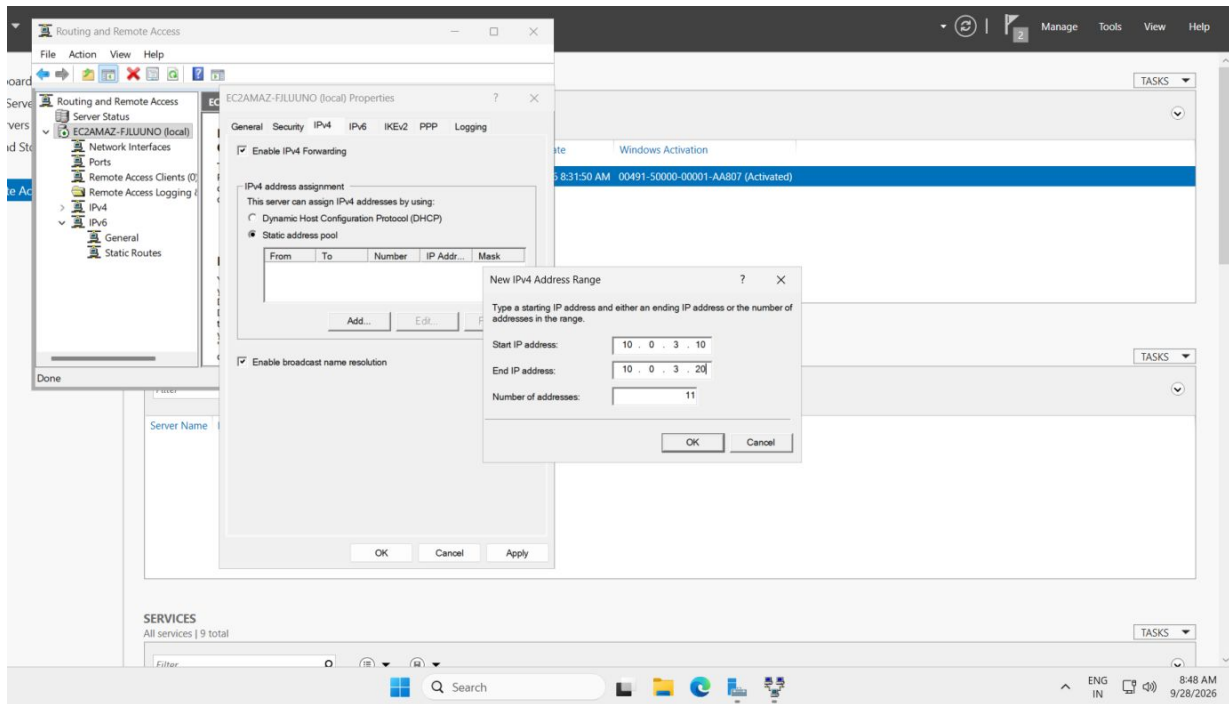


Fig 45. Add new IPv4 address range (start 10.0.3.10, end 10.0.3.20).

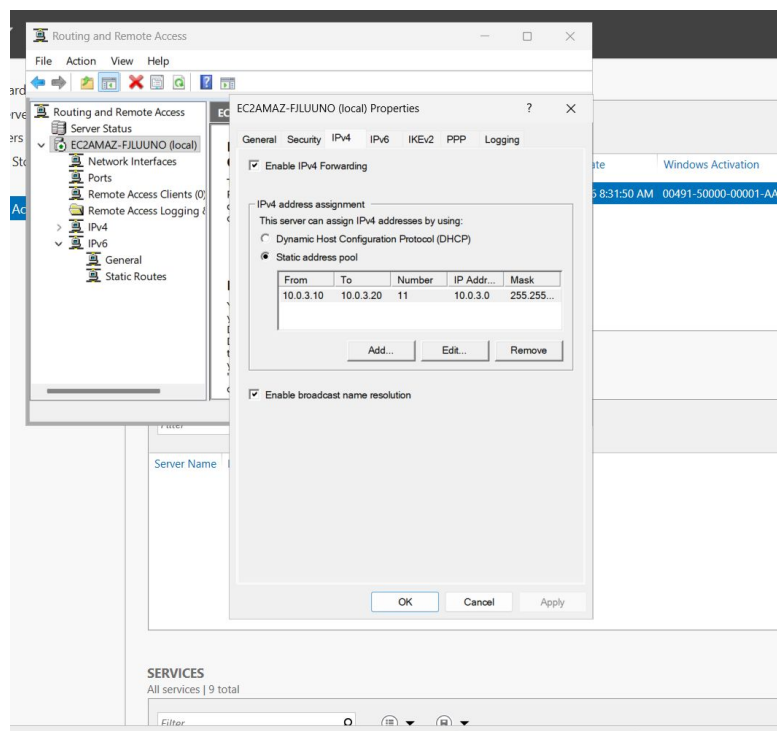


Fig 46. Static address pool saved.

Part 4 - VPN user, NAT, security groups

Step 4.1 - Create the VPN user

Open **Tools -> Computer Management -> Local Users and Groups -> Users -> New User**. Username used: **vpnserver**. Set a password and untick 'User must change password at next login'.

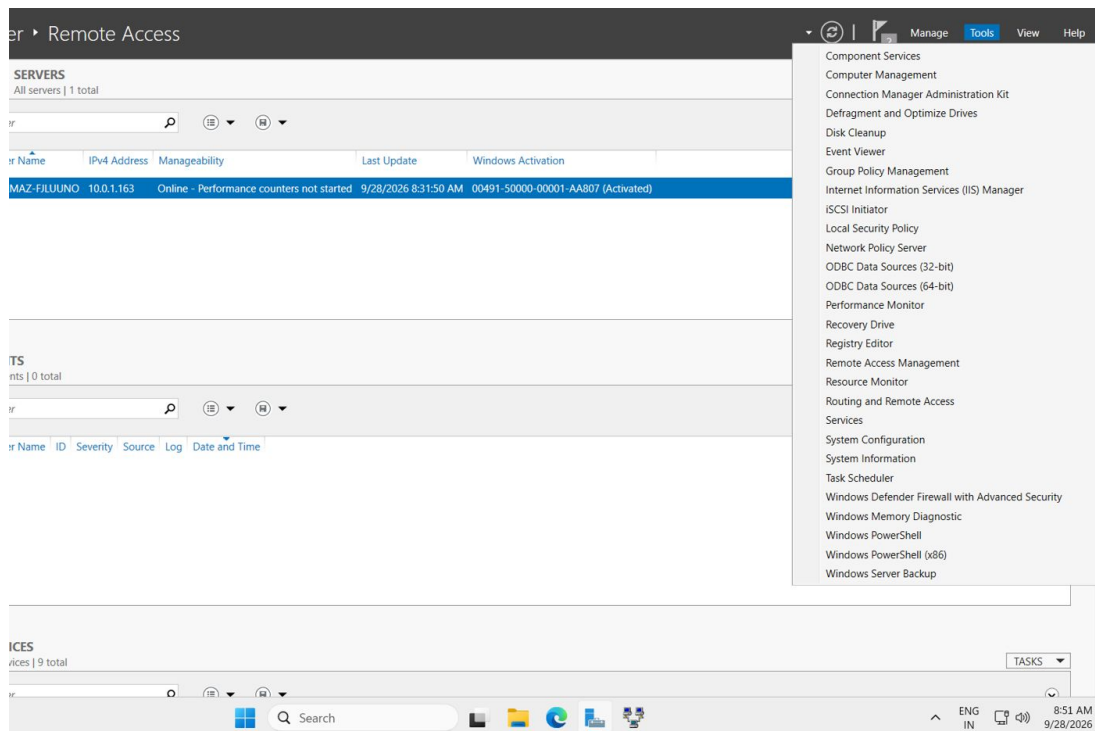


Fig 47. Server Manager -> Tools menu to open Computer Management.

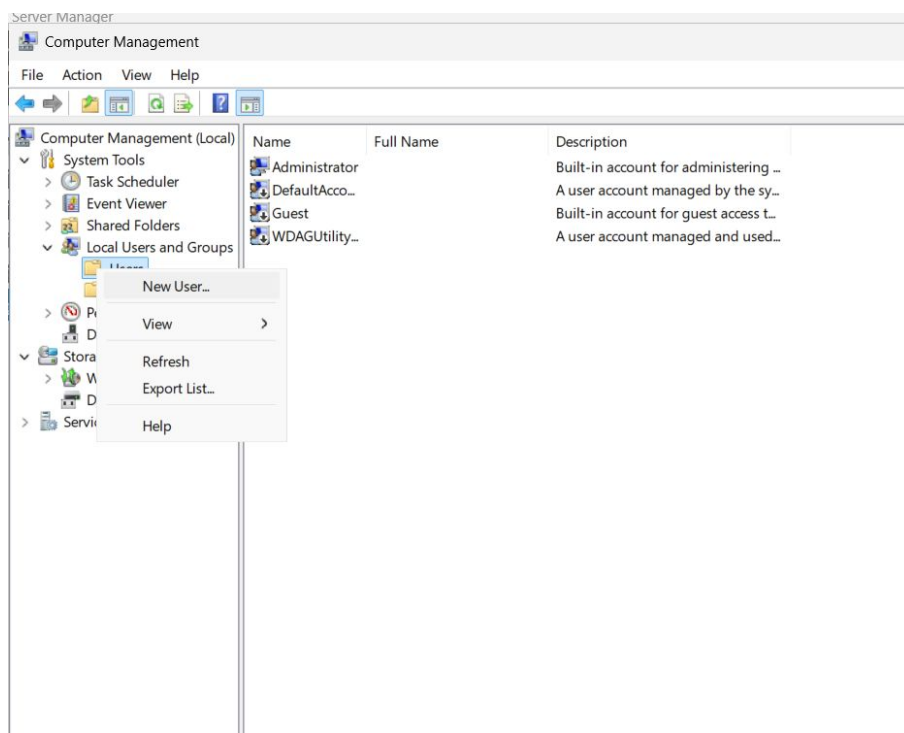


Fig 48. Users -> right click -> New User.

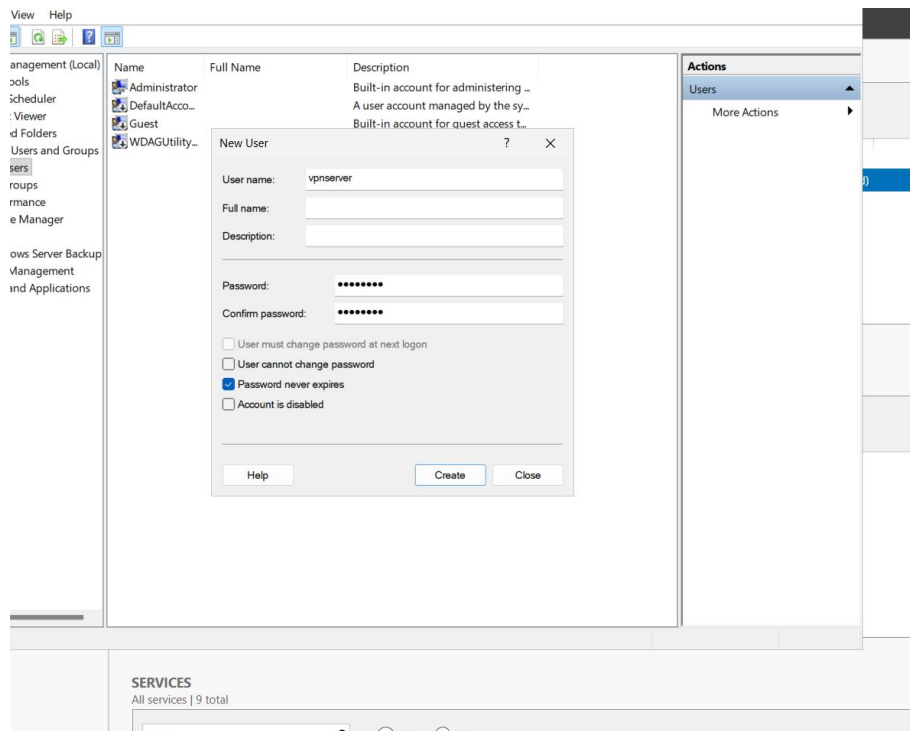


Fig 49. New user 'vpnservr' with password.

Step 4.2 - Allow dial-in (VPN) access

User Properties -> **Dial-in** tab -> Network Access Permission -> **Allow access** -> Apply.

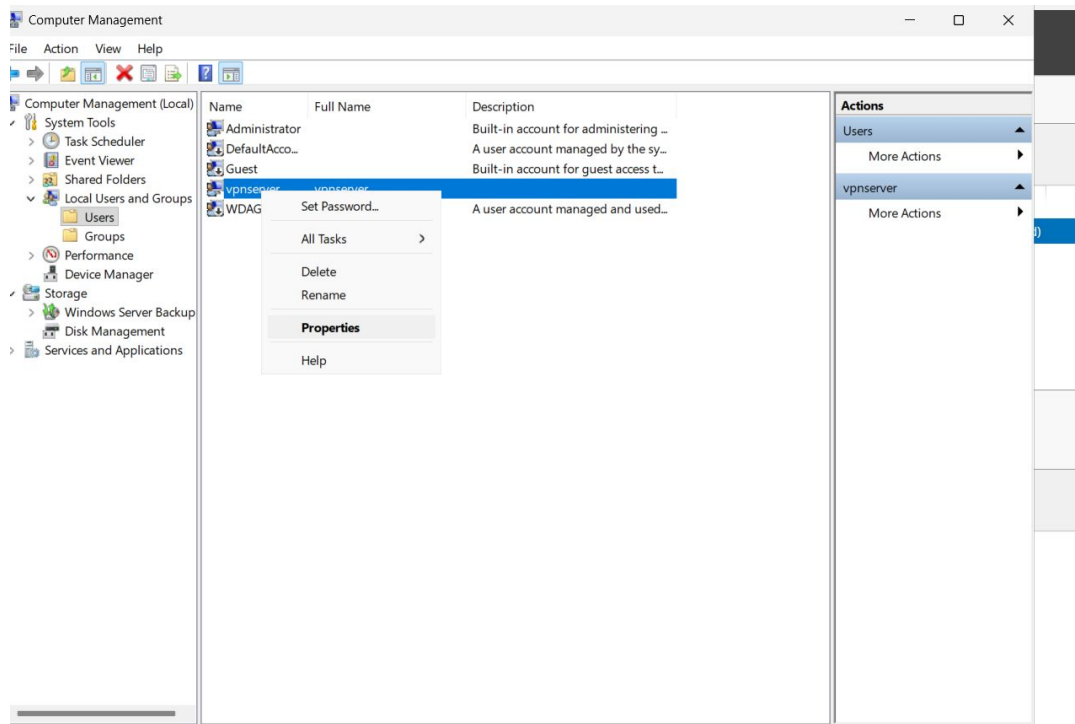


Fig 50. Right-click user -> Properties.

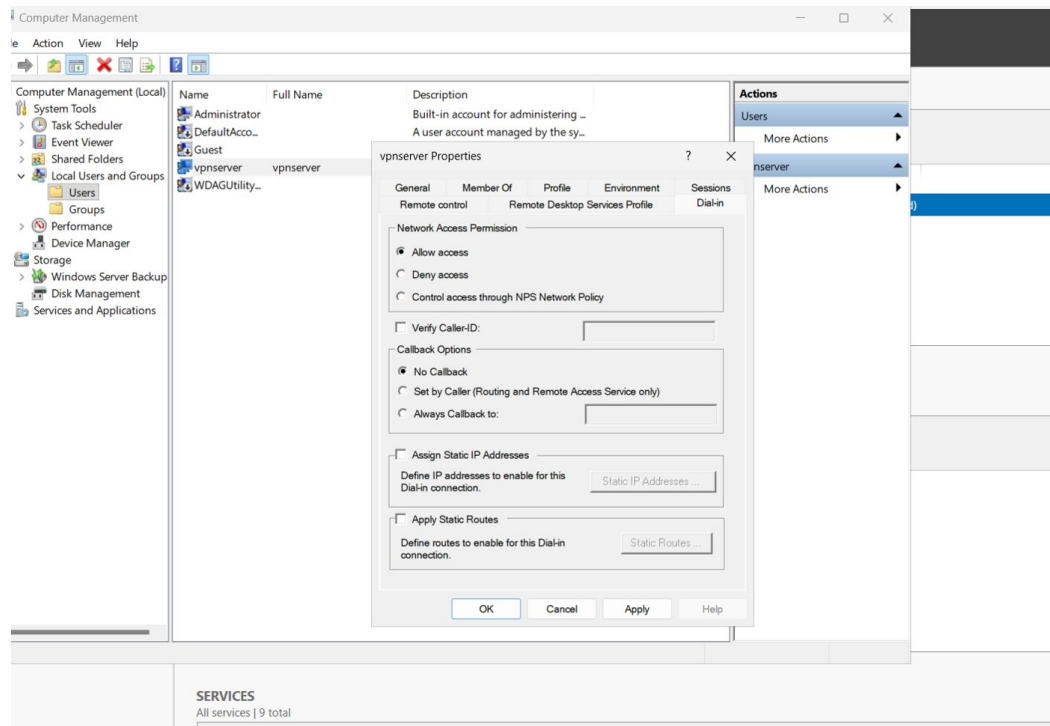


Fig 51. Dial-in tab: Allow access selected.

Step 4.3 - Restart the RRAS service

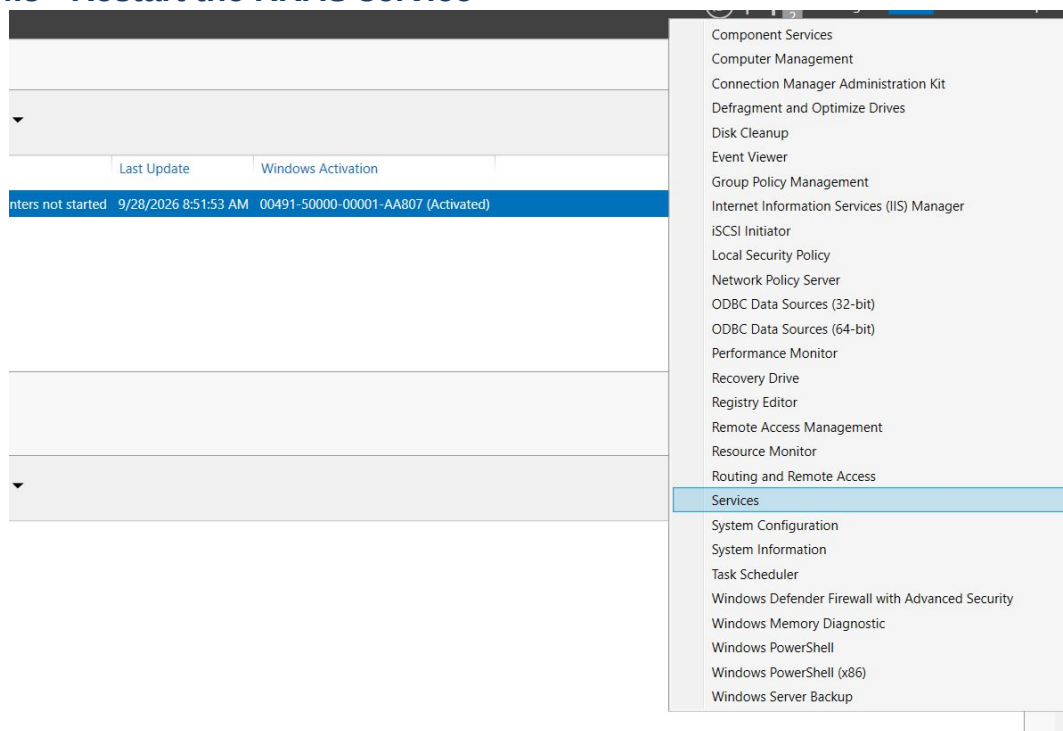


Fig 52. Tools -> Services.

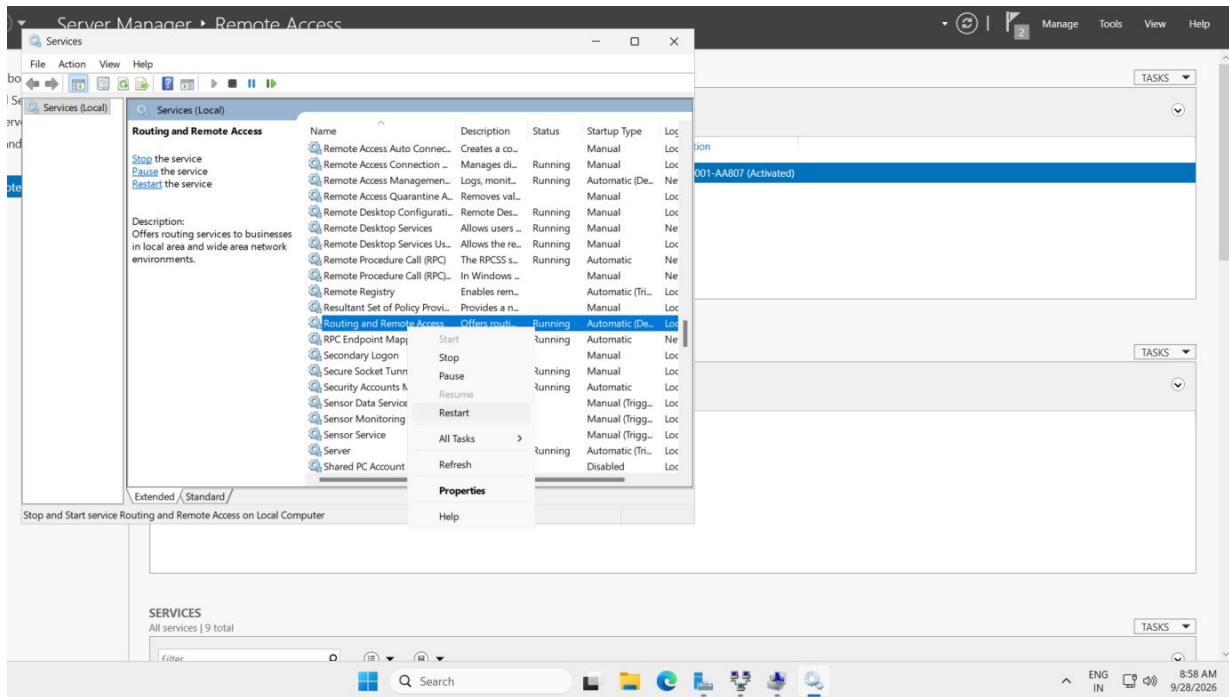


Fig 53. Routing and Remote Access service - Restart it so all changes apply.

Step 4.4 - NAT (so VPN clients can reach the VPC and internet)

In RRAS: **IPv4 -> General -> New Routing Protocol -> NAT**. Then add a **New Interface** (Ethernet) and mark it **Public interface connected to the Internet** with 'Enable NAT on this interface'. With NAT, traffic from the VPN client leaves the VPN server using the server's own IP (10.0.1.163), so the private EC2 accepts it easily.

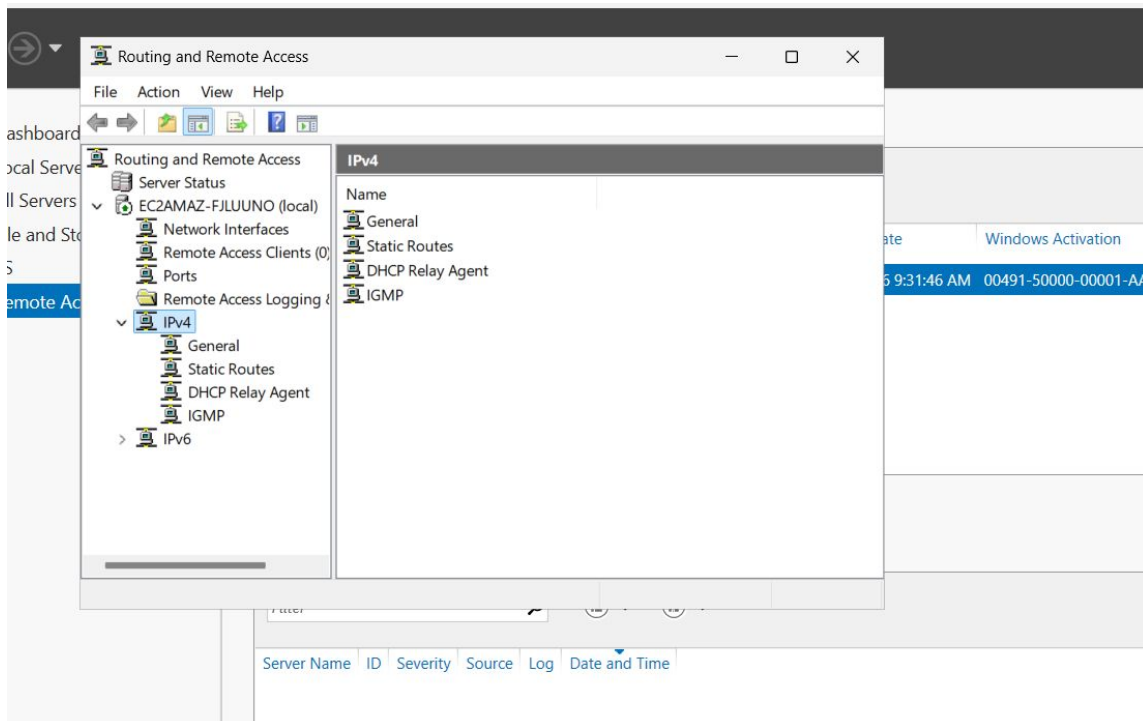


Fig 54. RRAS console - IPv4 tree.

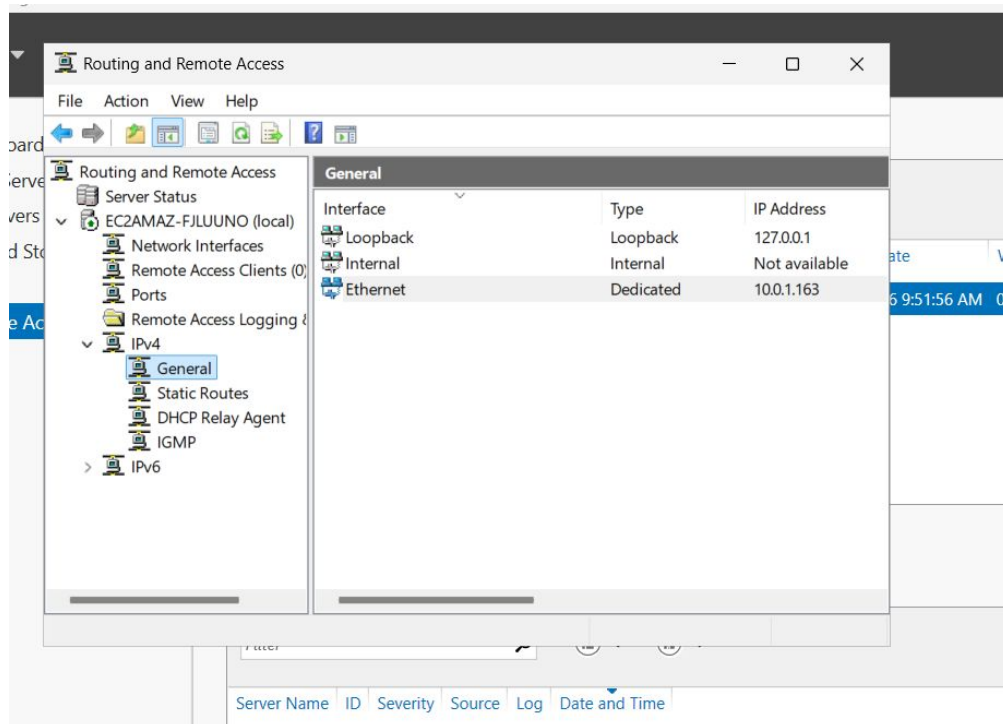


Fig 55. IPv4 -> General shows the Ethernet interface (10.0.1.163).

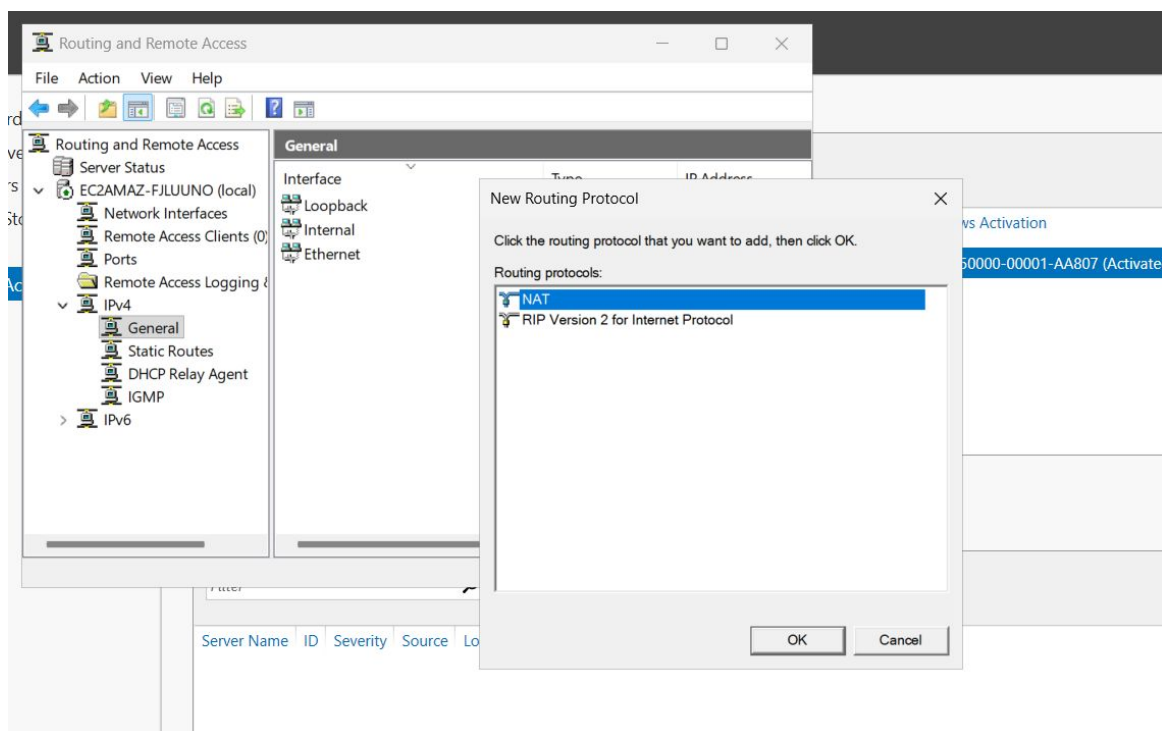


Fig 56. New Routing Protocol - select Network Address Translation (NAT).

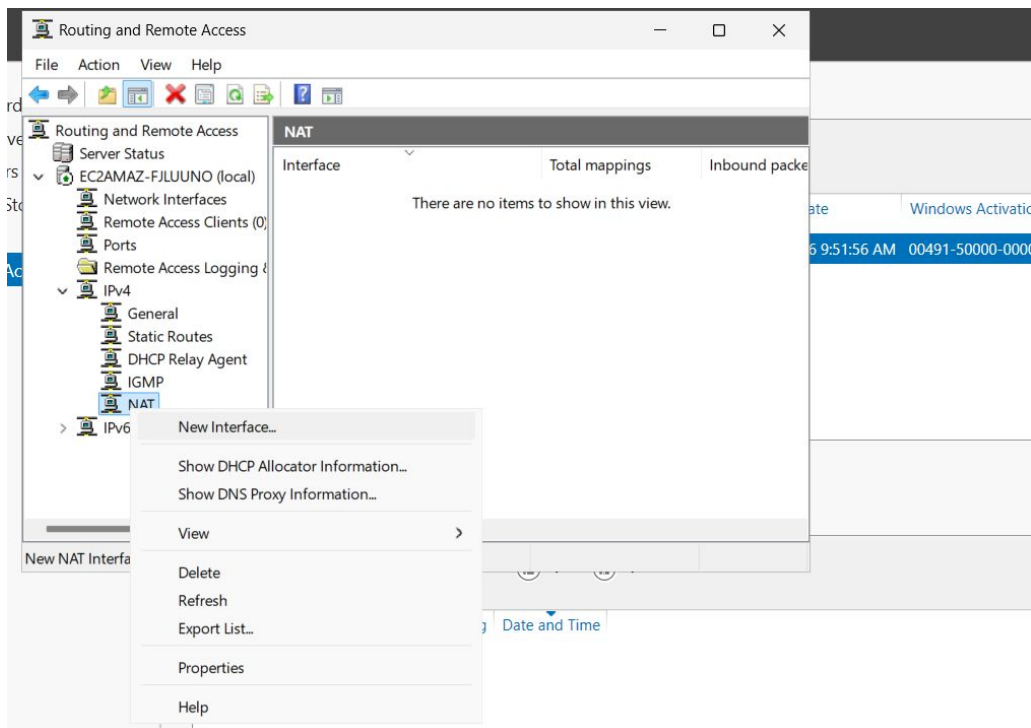


Fig 57. NAT added - right-click -> New Interface.

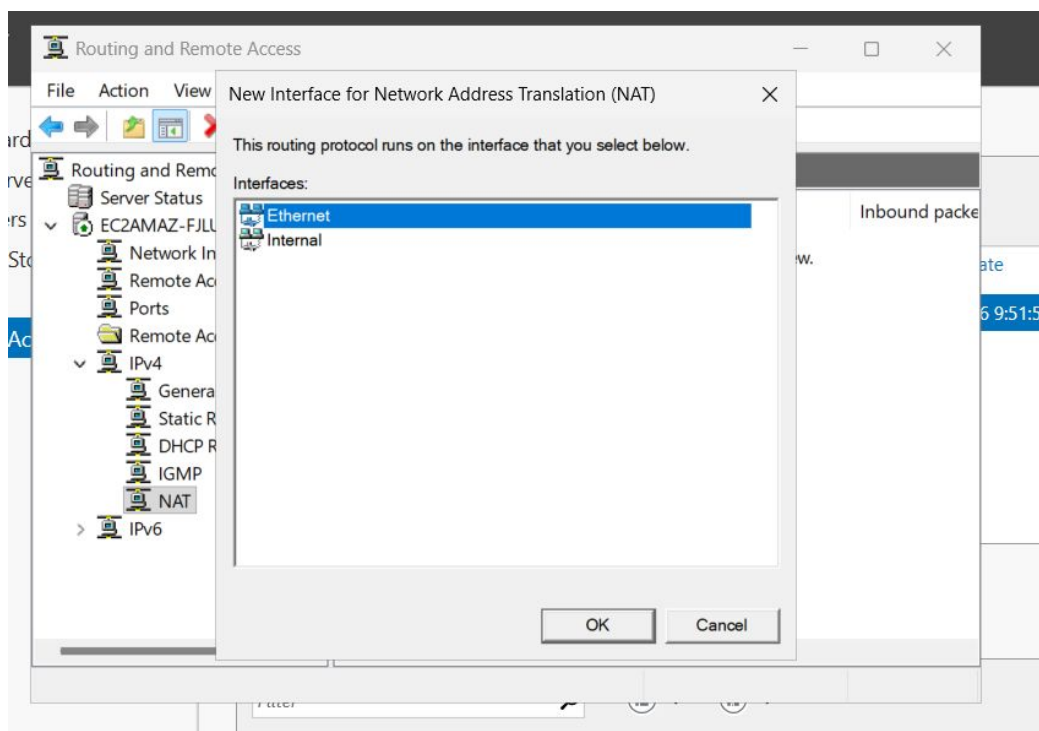


Fig 58. Choose the Ethernet interface for NAT.

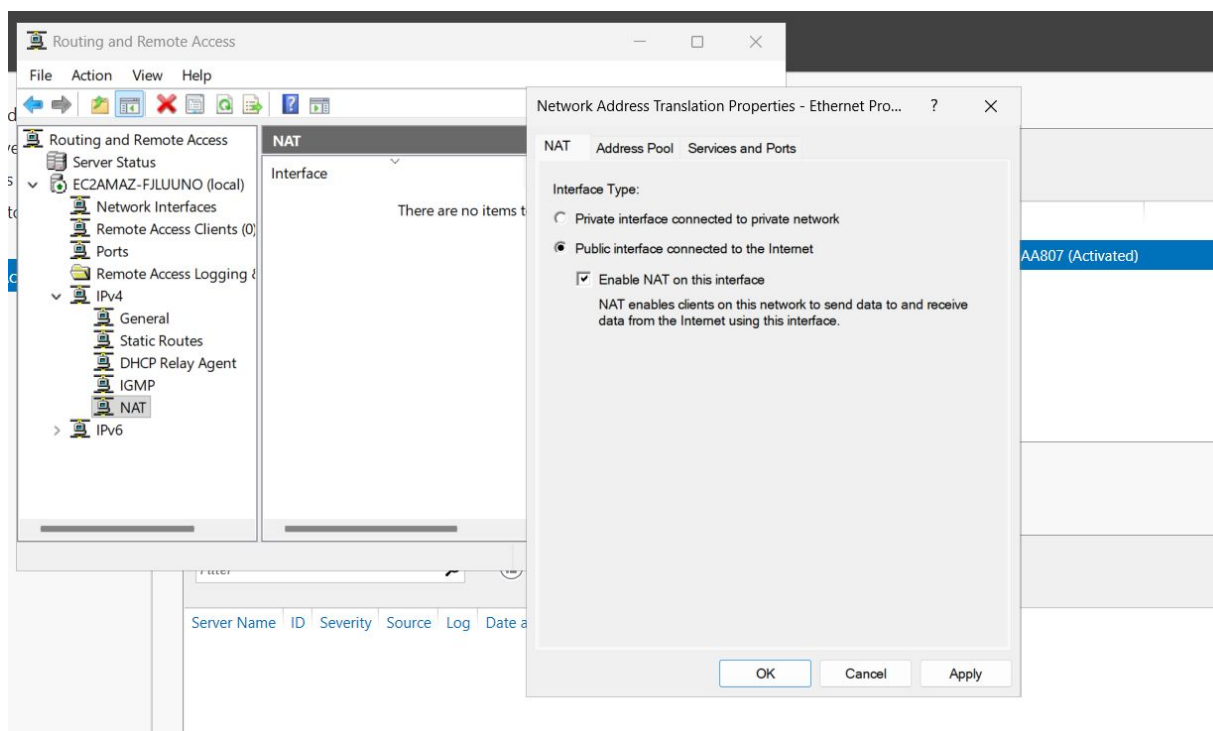


Fig 59. NAT properties: Public interface connected to the Internet, Enable NAT on this interface ticked.

Step 4.5 - AWS security group for the VPN server

On the VPN server's security group add these inbound rules:

Type	Protocol / Port	Why
Custom TCP	TCP 1723	PPTP control channel
Custom Protocol	GRE (Protocol 47)	PPTP data tunnel
RDP	TCP 3389	Admin RDP to the VPN server
All ICMP - IPv4 (optional)	ICMP	Allow ping for testing

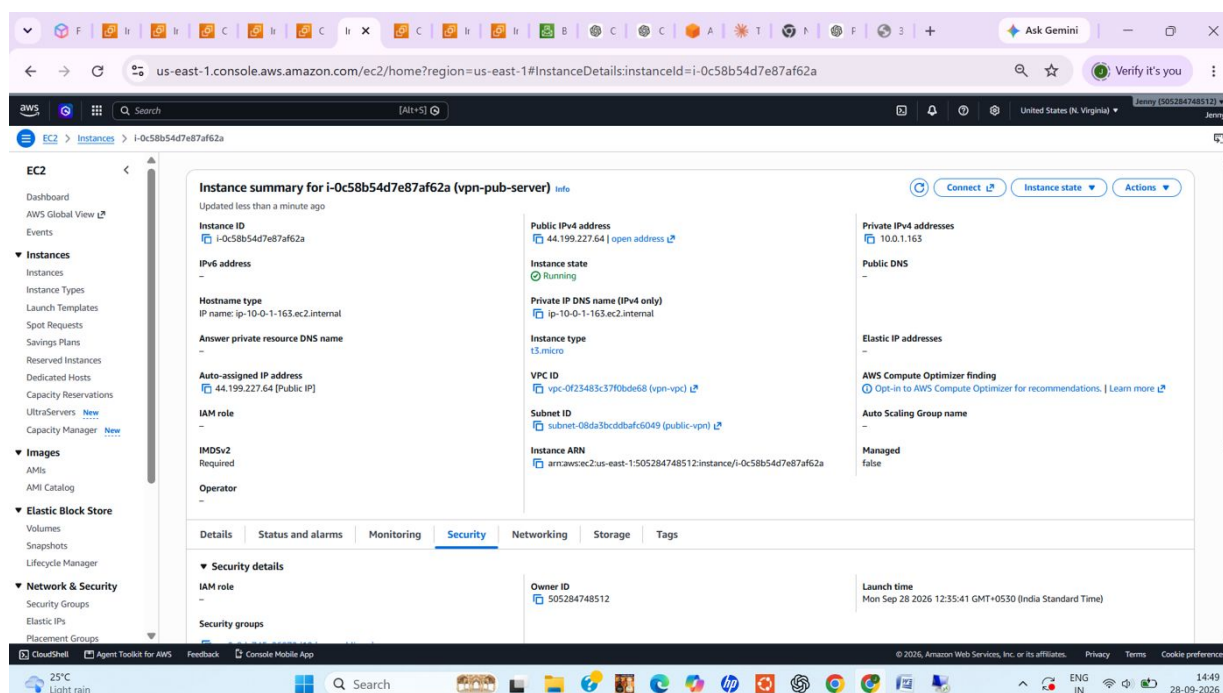


Fig 60. VPN server instance details -> Security tab.

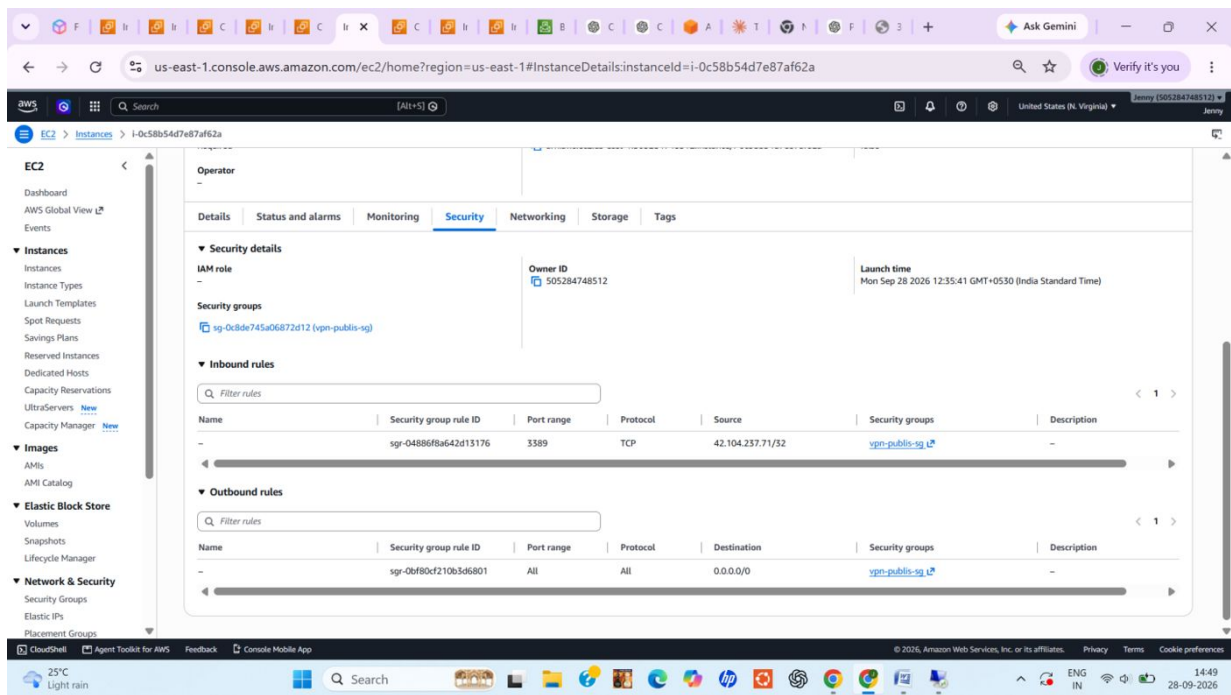


Fig 61. Security group inbound rules before editing.

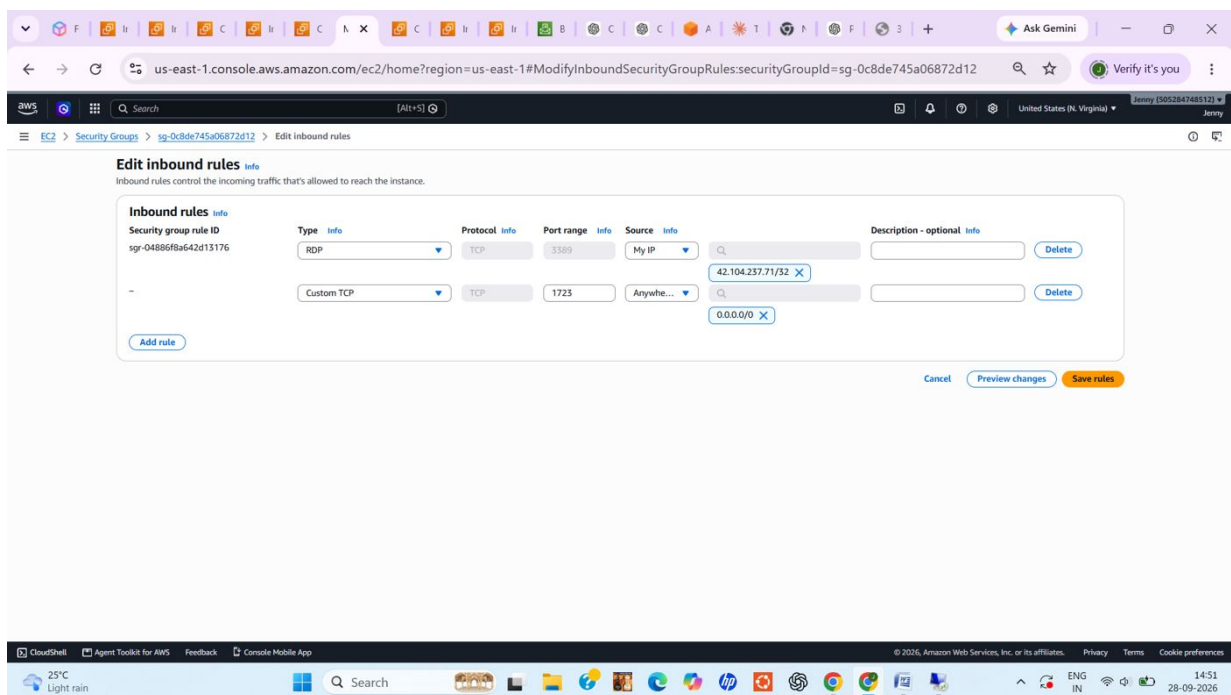


Fig 62. Edit inbound rules - Custom TCP 1723 added.

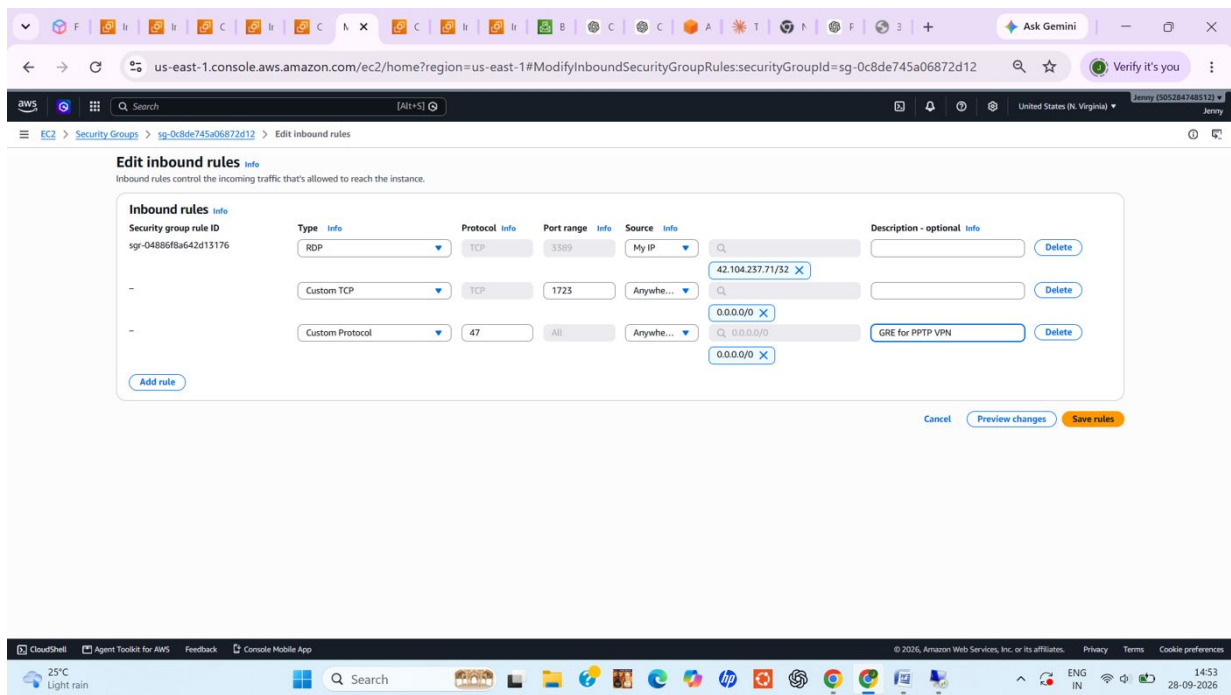


Fig 63. Custom Protocol 47 (GRE) added for PPTP.

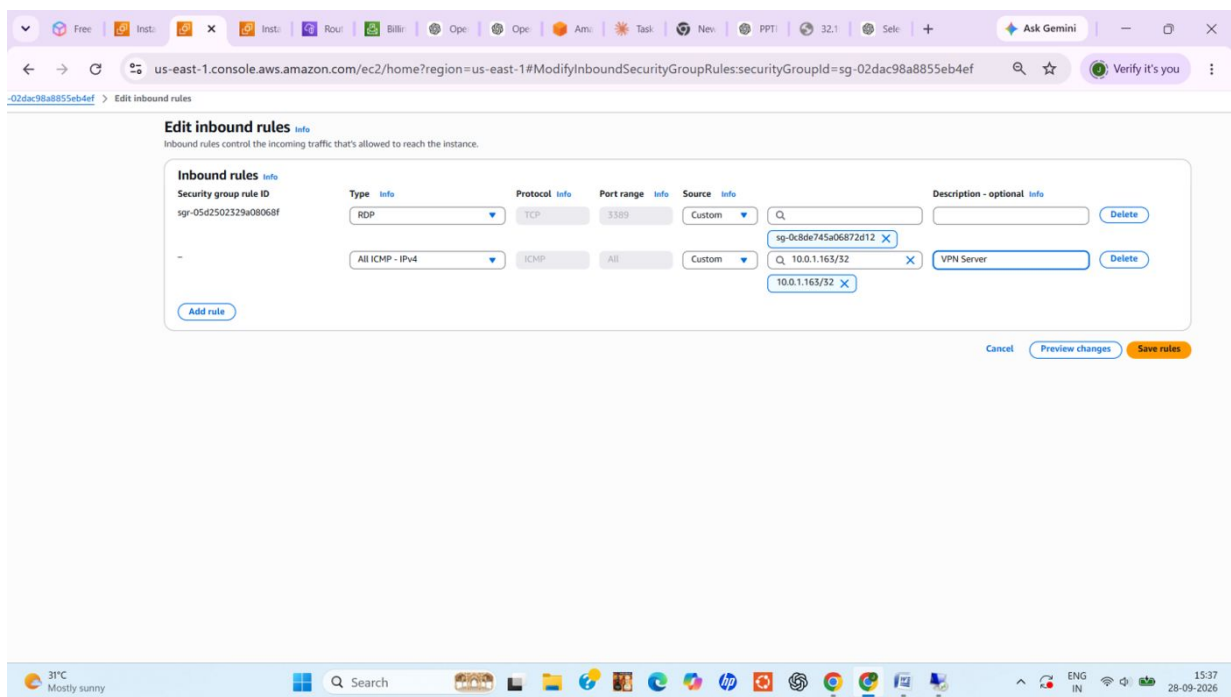


Fig 64. Private server security group - RDP allowed from the VPN server security group / VPC range.

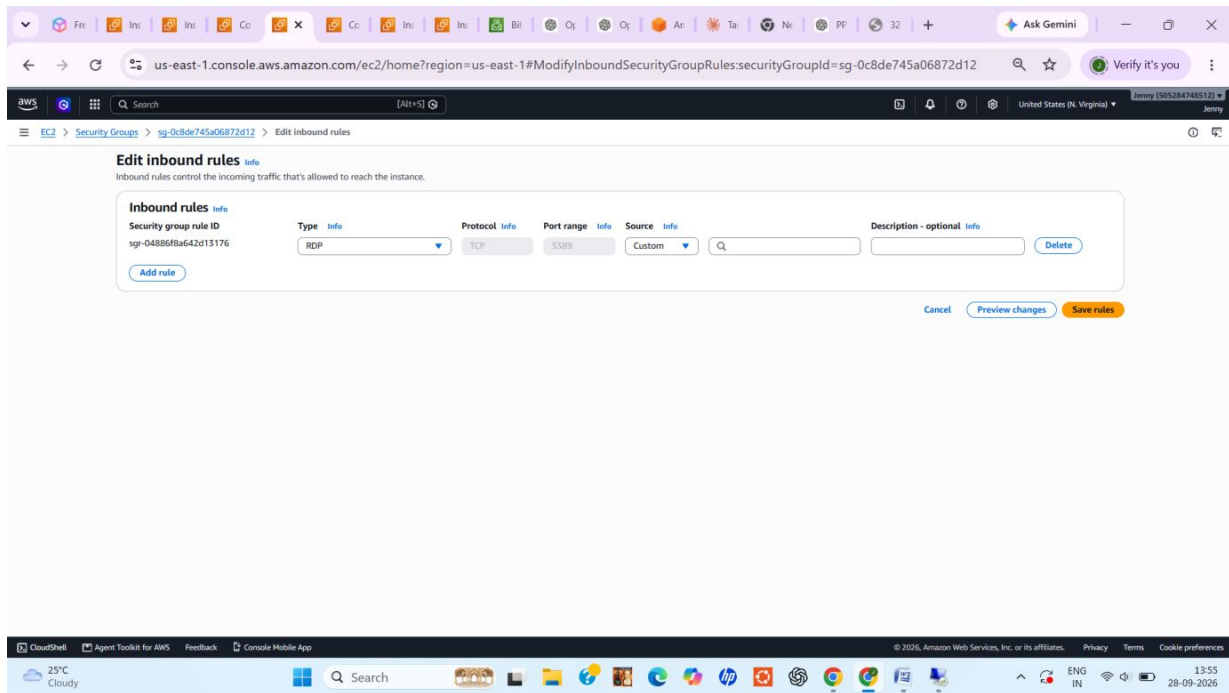


Fig 65. Optional ICMP rule to allow ping.

Step 4.6 - Disable Source/Destination check on the VPN server

An EC2 normally drops traffic that is not addressed to or from itself. A router/VPN server forwards other machines' traffic, so this check must be **stopped**: **Actions -> Networking -> Change source/destination check -> Stop**.

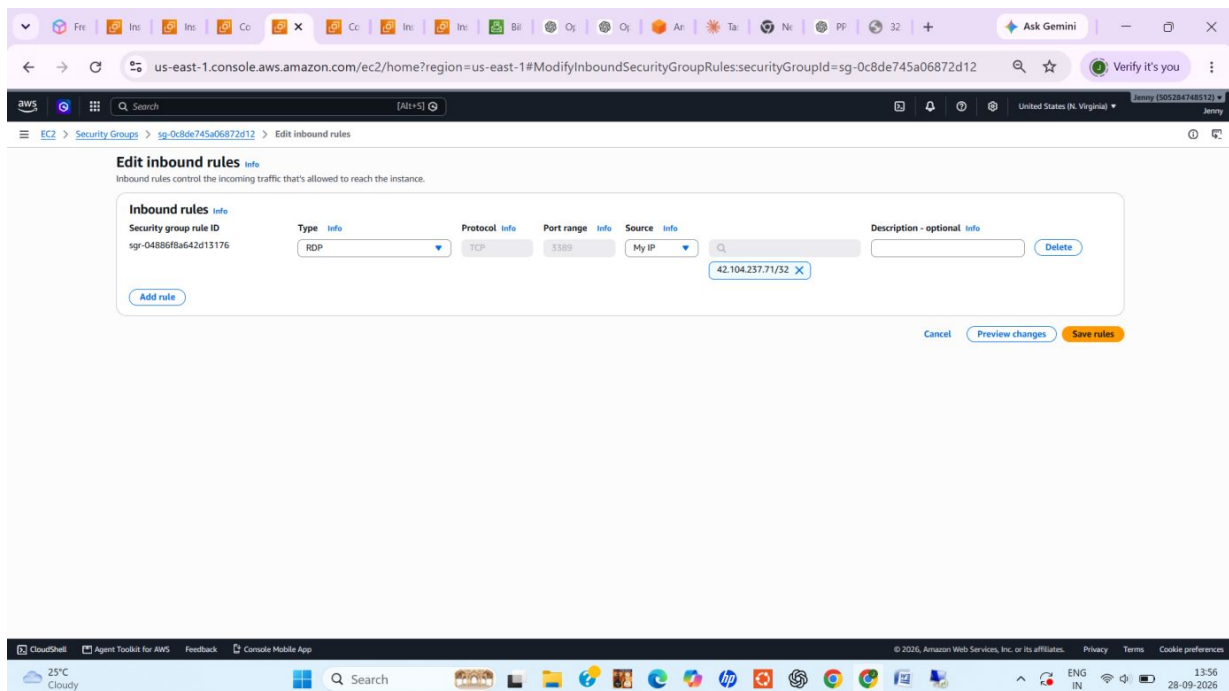


Fig 66. Security group changes saved.

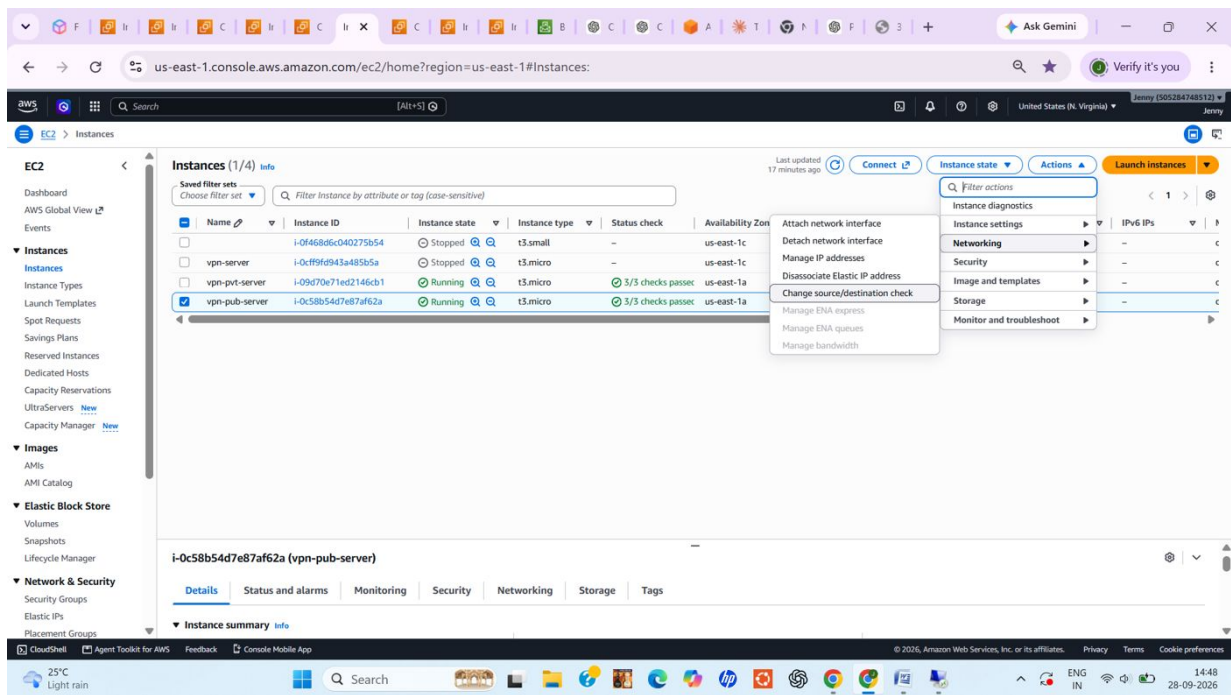


Fig 67. Actions -> Networking -> Change source/destination check.

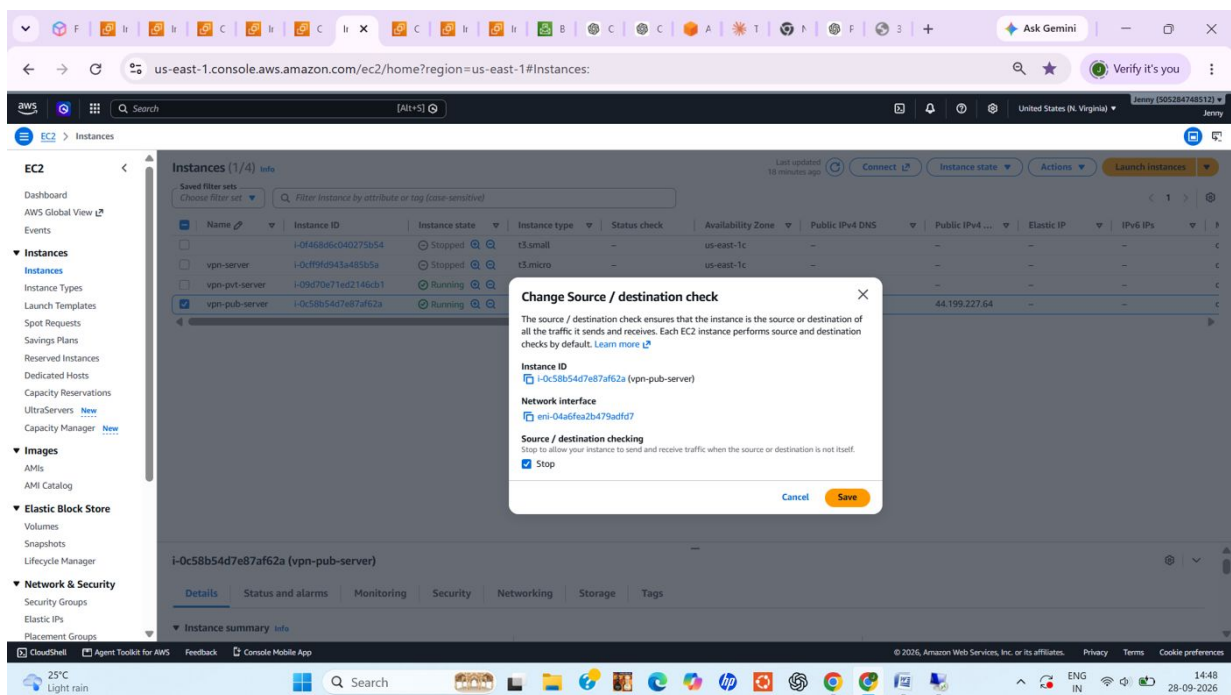


Fig 68. Tick 'Stop' and Save.

Part 5 - Laptop VPN, testing and final RDP

Step 5.1 - Create the VPN on the Windows laptop

The VPN is configured on the **laptop** (not inside the EC2 RDP session): **Settings -> Network & Internet -> VPN -> Add VPN**.

Field	Value
VPN provider	Windows (built-in)
Connection name	aws-vpn
Server name or address	44.199.227.64 (VPN server public IP)
VPN type	Point to Point Tunneling Protocol (PPTP)
Sign-in type	Username and password
Username / Password	vpnservice / (the password you set)

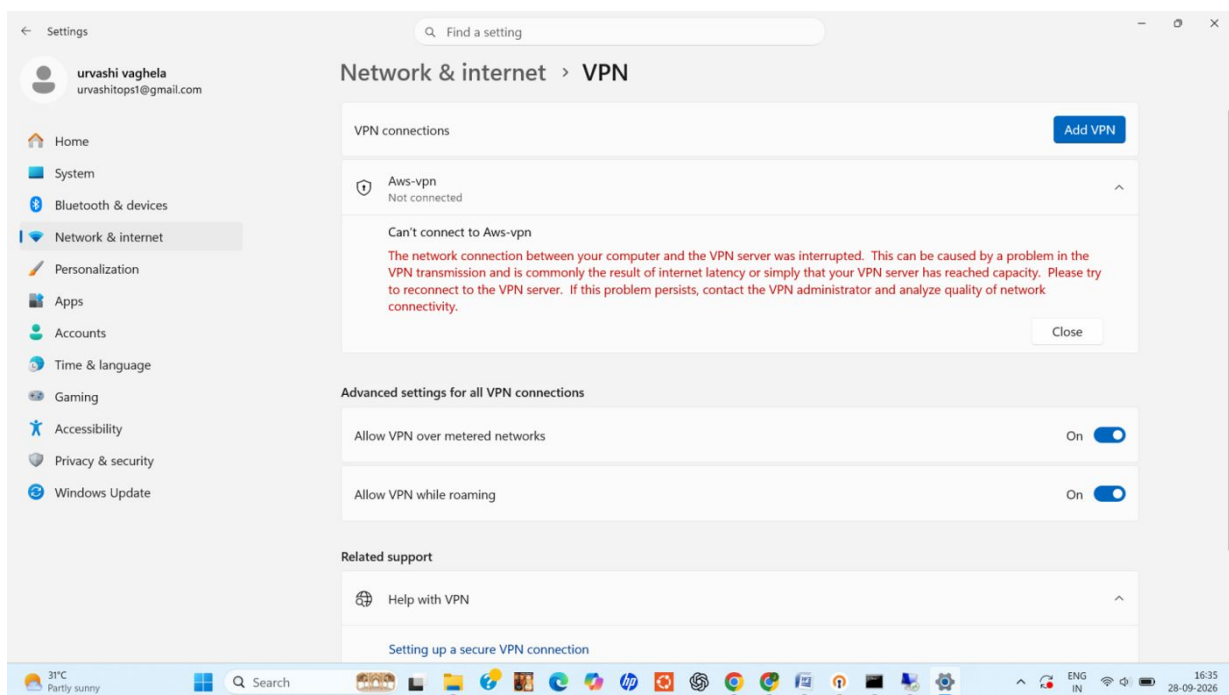


Fig 69. Laptop VPN page - first attempt showed an error because the network/ports were not ready.

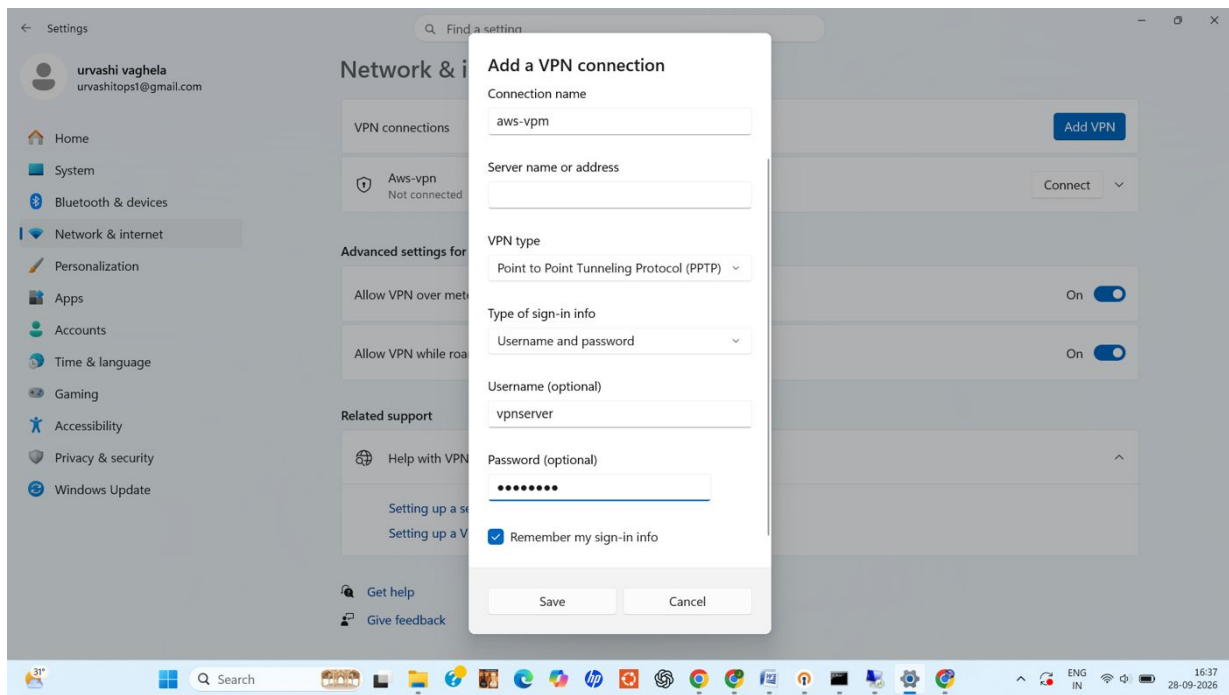


Fig 70. Add a VPN connection form filled with the values above.

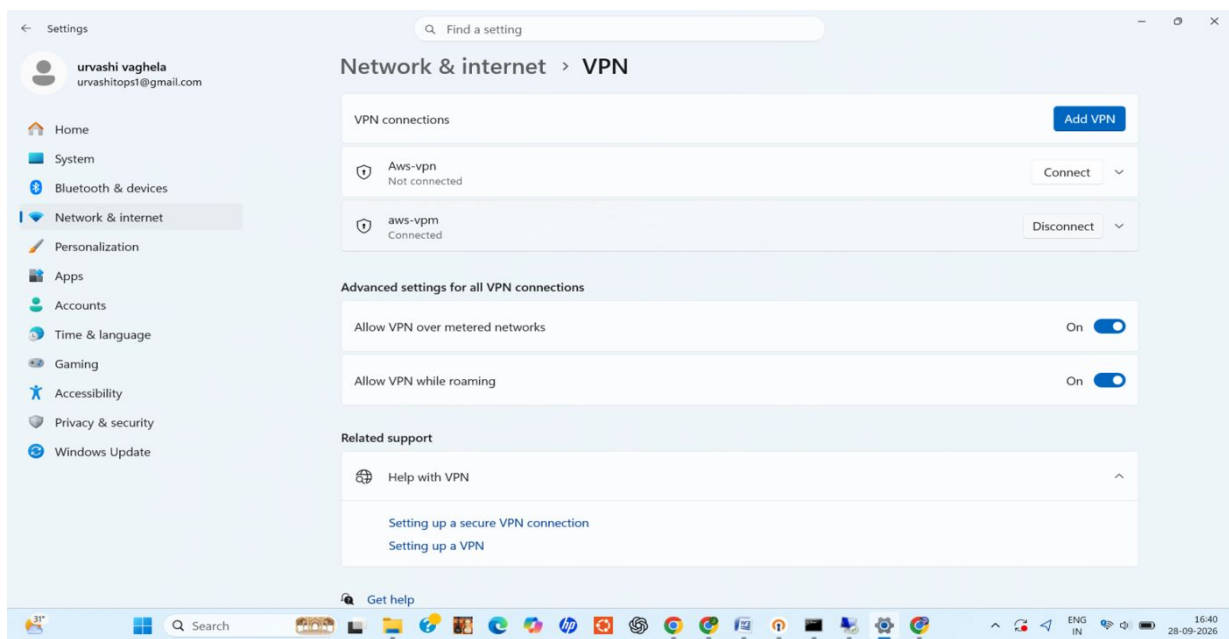


Fig 71. VPN list - click Connect. It shows Connected.

Step 5.2 - Verify the VPN IP with ipconfig

On the laptop open Command Prompt and run **ipconfig**. The **PPP adapter aws-vpn** shows IPv4 **10.0.3.12**, mask 255.255.255.255. This proves the VPN server gave a client IP from its pool.

ipconfig

```
Select C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\URVASHI>ipconfig

Windows IP Configuration

PPP adapter aws-vpm:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 10.0.3.12
    Subnet Mask . . . . . : 255.255.255.255
    Default Gateway . . . . . : 0.0.0.0

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:8fef:e73c:fa18:a2f:1104:7610
    Temporary IPv6 Address. . . . . : 2401:4900:8fef:e73c:a8e6:7c06:98aa:f7ad
    Link-local IPv6 Address . . . . . : fe80::ea5:3957:f21:c7dd64
    IPv4 Address. . . . . : 192.168.1.139
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1033:75ff:feed:45b3%4
                                192.168.1.1

C:\Users\URVASHI>
```

Fig 72. ipconfig on the laptop - PPP adapter shows the VPN IP 10.0.3.12.

Step 5.3 - Ping test (and why it timed out)

```
ping 10.0.2.45
ping 10.0.1.163
```

Both replied **Request timed out**. This does **not** mean RDP is broken - Windows Firewall and security groups usually block ICMP (ping). So the real service port was tested instead.

Step 5.4 - Test the actual RDP port

```
Test-NetConnection 10.0.2.45 -Port 3389
```

Result: **InterfaceAlias = aws-vpm, SourceAddress = 10.0.3.12, TcpTestSucceeded = True**. This is the most important proof - traffic goes from the laptop's VPN IP to the private EC2 on TCP 3389.

```
PT
C:\WINDOWS\system32\cmd.exe - powershell

Pinging 10.0.1.163 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.1.163:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\URVASHI>powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\URVASHI> Test-NetConnection 10.0.2.45 -Port 3389
PS C:\Users\URVASHI> Test-NetConnection 10.0.2.45 -Port 3389

ComputerName           : 10.0.2.45
RemoteAddress          : 10.0.2.45
RemotePort             : 3389
InterfaceAlias         : aws-vpm
SourceAddress          : 10.0.3.12
TcpTestSucceeded       : True
```

Fig 73. Ping timed out, but Test-NetConnection to 10.0.2.45 port 3389 says TcpTestSucceeded : True.

Step 5.5 - Final RDP to the private EC2

Keep the VPN connected. Press **Win + R -> mstsc**, enter **10.0.2.45** and connect with the private EC2's Windows credentials.

- Do NOT use 44.199.227.64 (VPN server public IP)
- Do NOT use 10.0.1.163 (VPN server private IP)
- Do NOT use 10.0.3.12 (laptop's own VPN IP)
- Correct destination: **10.0.2.45**

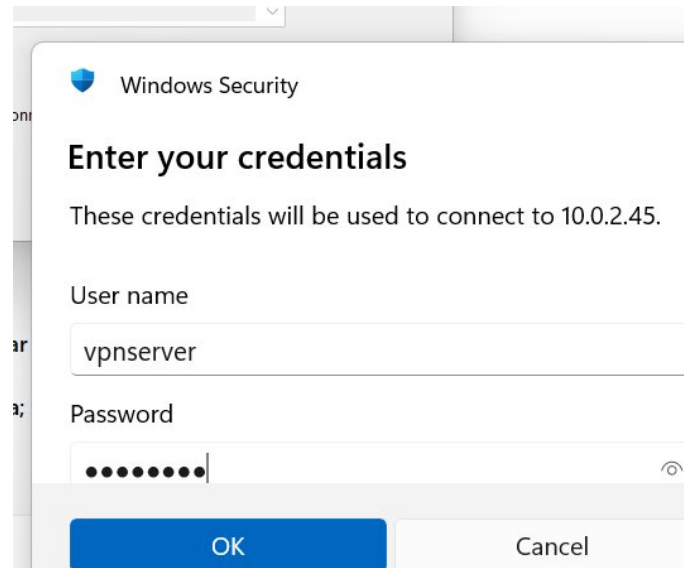


Fig 74. Remote Desktop asking credentials for 10.0.2.45.

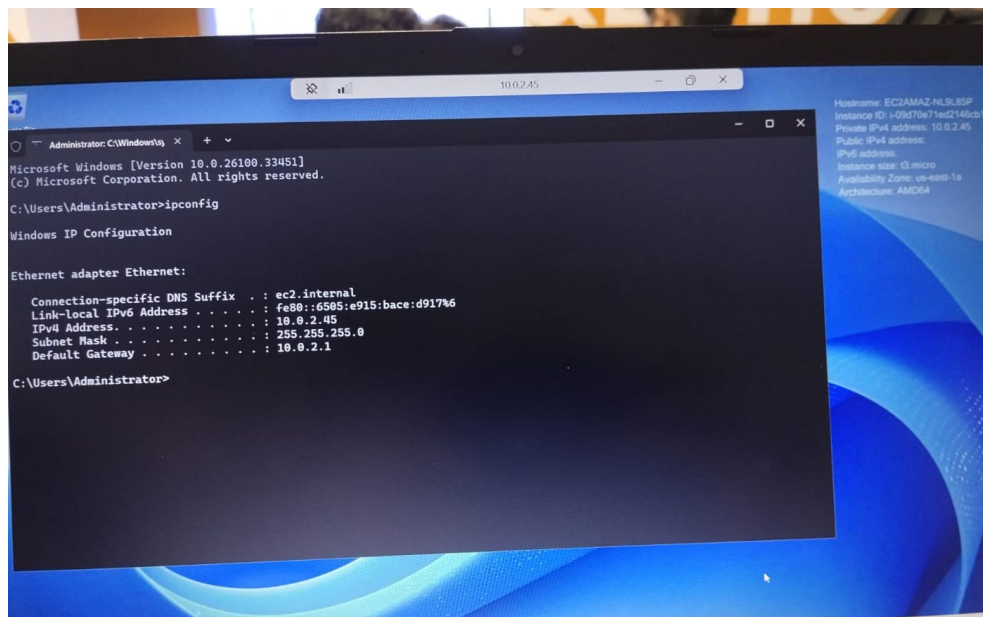


Fig 75. RDP session is open - ipconfig inside it shows IPv4 10.0.2.45 (private EC2, no public IP).

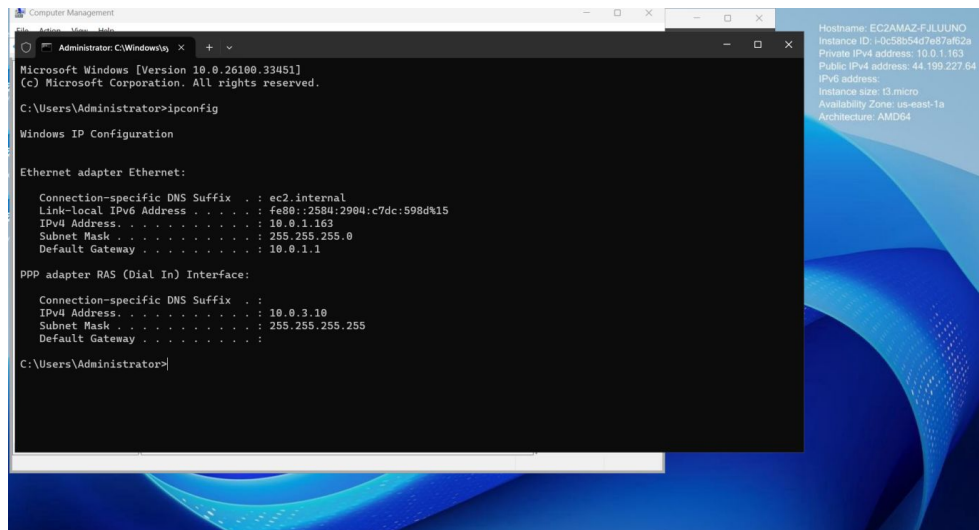
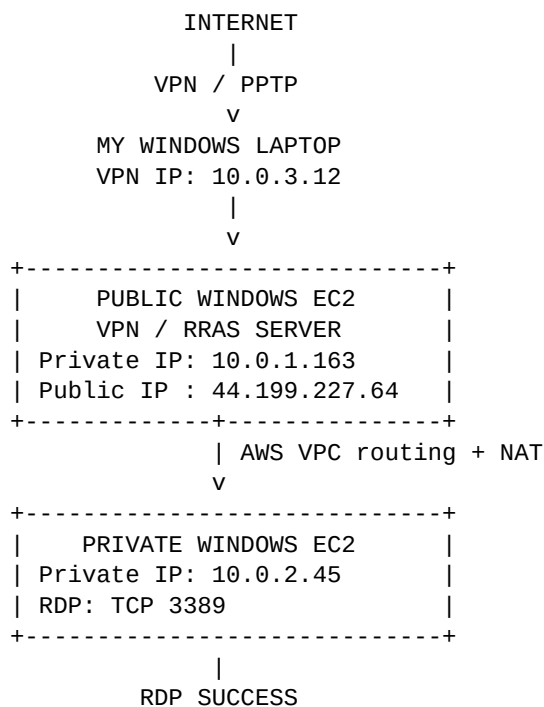


Fig 76. ipconfig on the VPN server - Ethernet 10.0.1.163 and a PPP (dial-in) adapter for the connected VPN client.

6. Final result

The laptop connected to the VPN, received **10.0.3.12**, passed **TcpTestSucceeded : True** on port 3389, and opened an RDP session to the private Windows EC2 **10.0.2.45** without any public IP on that server.



7. Real-life use

Companies keep important servers (databases, admin servers, file servers, internal apps) in **private subnets** with no public IP so nobody on the internet can even see them. Employees still need to reach them, and a VPN is the safe doorway.

Real situation	How this setup is used
Work from home	Employee connects VPN from home and then RDP/SSH to office or cloud servers by private IP.
Admin access to private servers	Admins reach DB / application servers without giving them public IPs - much smaller attack surface.
Replacement for a bastion / jump host	One VPN entry point instead of exposing RDP on many servers.
Contractors / vendors	Give a temporary VPN user with dial-in permission, remove it when the work ends.
Dev / test environments	Developers reach private test servers and internal websites securely.
Branch office / small business	A Windows server as a low-cost VPN gateway for a small team.

Note for production: PPTP is old and considered weak. It is perfect for learning because it is built into Windows and easy to set up. In real companies use stronger options: **SSTP / IKEv2 / OpenVPN, AWS Client VPN, or AWS Site-to-Site VPN**, plus MFA and least-privilege security groups.

8. Short theory (easy words)

Term	Meaning
VPN	An encrypted tunnel over the internet that makes your laptop behave as if it is inside the remote network.
RRAS	Routing and Remote Access Service - the Windows feature that turns a server into a VPN server and router.
PPTP	Simple VPN protocol built into Windows. Uses TCP 1723 and GRE (protocol 47). Easy but not very secure.
NAT	Replaces the source IP of packets so traffic looks like it comes from the server itself (10.0.1.163).
Routing	Deciding where a packet should go next (VPN pool -> VPC subnets).
Public IP	Reachable from the internet (44.199.227.64).
Private IP	Reachable only inside the VPC / VPN (10.0.1.163, 10.0.2.45).
RDP	Remote Desktop Protocol - graphical login to a Windows machine.
TCP 3389	The port on which Windows listens for RDP.
Source/Dest check	EC2 safety feature; must be disabled on any instance that forwards other machines' traffic.

9. Commands and IP summary

Task	Command / action
Check VPN IP	ipconfig
Ping (may time out)	ping 10.0.2.45
Test RDP port	Test-NetConnection 10.0.2.45 -Port 3389 (expect TcpTestSucceeded : True)
Open Remote Desktop	Win + R -> mstsc -> 10.0.2.45

10. Common problems and fixes

Problem	Fix
VPN error 'network connection between your computer and the VPN server was interrupted' (Fig 69)	Open TCP 1723 and GRE (protocol 47) in the VPN server's security group; check public IP is correct; restart the RRAS service.
Authentication failed	Check username/password; user Dial-in tab must be 'Allow access'.
VPN connects but no IP from 10.0.3.x	Check the static address pool (10.0.3.10-10.0.3.20) and restart RRAS.
Ping timeout but everything else fine	Normal - ICMP blocked. Use Test-NetConnection on port 3389.
TcpTestSucceeded : False	Private EC2 security group must allow RDP (3389) from the VPN server SG / VPC range; source/dest check must be Stopped; NAT enabled.
RDP asks wrong credentials	Use the private EC2's own Administrator password (decrypt with its key pair), not the VPN user's.